

Electric Vehicles:

23 July 2016
Clemens

Introduction: An electric vehicle is a vehicle that uses one or more electric motors for propulsion instead of utilizing an ICE.

Basic Description of Working Principle:

Electrical Energy $\rightarrow$ Electrical Motor $\rightarrow$ Mechanical Energy

ICE $\rightarrow$ Chemical Energy $\rightarrow$ Heat $\rightarrow$ Mechanical $\rightarrow$ Rotational motion

First Component of EV :-

1) Battery - Lead Acid, Lithium Ion

1900-1920 - EV very popular (Golden Era) after 1920 - fuel pressure and speed were built. - Shifted to ICE - EVs got drawn back.

function - To store electrical energy. - supply DC power to motor - determines range.

Different EVs - figure out battery pack size of EVs

Total Mass - 4500 kg

2) Electric Motor - IM, BLDC, PMSM

Power loss - It's the electrical energy lost as heat in conductors and transformers

- Causes:
- 1) High charging current
 - 2) Long cable length (Higher impedance drop more loss)
 - 3) Overloaded conductors
 - 4) Harmonic currents

Effect - Increased electricity cost

- 1) Heating of cables and transformers
- 3) Reduced overall efficiency

Solutions -

- 1) Reduced feeder length where possible
- 2) Improve power factor
- 3) Use large conductor size
- 4) Install charging station closer to substation

Overloading of transformer - It occurs when the load increases the 11% rated capacity

Effects -

- Faster insulation aging
- Overheating
- Reduced life of 11%
- Increased maintenance

Solutions -

- 1) Using smart charging
- 2) Load management
- 3) Upgrade 11%
- 4) Install meter 11% (distribution 11%)

The Power Quality degradation:

It refers to maintaining a clean stable and reliable supply of electrical power with correct V, f and waveform

- Causes -
- Harmonics, voltage sag, fluctuations, unbalanced loading
 - Poor power factor, unbalanced loading

Effects -

- Reduced equipment life
- Increased switching, sensitive electronic malfunction

Solution -

- Smart grid technology, Energy storage system, harmonic filters, demand response program, demand response can improve power monitoring

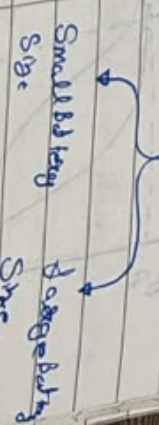
Basic Vehicle Parameters:

- 1) Engine Power
- 2) Range - capacity of fuel
- 3) Efficiency - How efficient the car is
- 4) Fixed Power - driving up or down a slope
- 5) Acceleration - Power of motion, slope, speed etc

- Fuel consumption
- Fuel (weight)
- Road condition
- Driving style (speed etc)
- driving cycle - Real drive cycle

St. August's

BEV



increase this for local community storage

Impact of modern drive train of power supplies

Drive train - Collection of component driving circuits

14/11/2026

Impact - +ve & -ve
Power supply - fixed.

17/11/26 EV charging simultaneous - 100kWh - 17/11/26

Voltage instability - If the condition in which the supply voltage changes from its normal value due to sudden or excessive loading.

5% change of deviation - fixed 415V - 399 - 441V.

Cause of voltage stability -

1) Sudden charging / load increase - It increases load which causes feeder voltage to drop.

2) Long distribution feeder - load near to T/L with power loss impedance and farther away will have high impedance.

Consumers located far from distribution will receive larger voltage drop.

3) Load distribution - Old and underused distribution lines and underused conductors and overloaded feeders can't maintain voltage during EV charging demand.

Fast DC charger - Rapid charging in charging power stations can produce sudden voltage fluctuations.

Effects - Reduced motor efficiency

2) Malfunction in electronic equipment

3) slower charging speed

fixes / improvements - 1) Smart Chargers -

2) Upgraded infrastructure

3) smart voltage regulators.

Harmonics - Causes - EV chargers contain power electronic converters

Effects - T/L overloading, cable overheating, additional power losses and reduced motor efficiency

Solution - Upgradation of actual passive filters.

High power factor chargers - voltage sag

Voltage sag - Causes - fast charging station switching on - creates immediate drop.

Sudden increase in EV charging load

Effects - Battery charging errors after

2) charger interruptions.

3) Solutions -

Solutions - 1) Soft charging starter

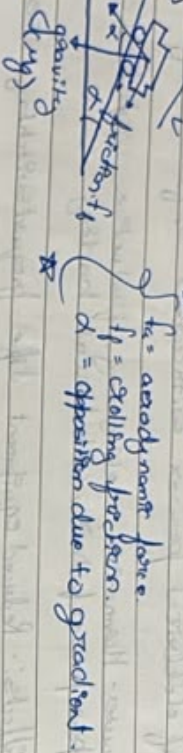
2) Dedicated feeders

3) Energy storage system.

4) Dynamic voltage restorers.

Dynamic Model for vehicle:

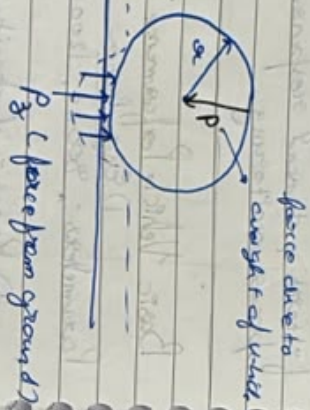
Along the second class of motion $\rightarrow \frac{dv}{dt} \propto P - F_{\text{res}} - F_{\text{drag}}$



To move the vehicle up the slope we have to produce force to exceed $f_r, f_g, \text{ and } f_d$.

1.) Rolling Resistance:

When Car is at standstill:



There can't be the same unbalanced loading condition as in the figure of a car. There will be hysteresis that will cause different deformation of tyre causing different path.

When tyre removing:

Tyres gets more deformed from the front to maintain a bit more friction to make the car stable and grippy.



Factors affecting rolling resistance:

- 1) Material of tyre - hard or soft.
- 2) Tyre structure - f_g
- 3) Tyre Temperature
- 4) Tyre inflation pressure - To reduce rolling resistance higher pressure.
- 5) Road roughness - Material of road - presence of liquid on road.

Acceleration of a vehicle

Using Newton's 2nd Law of motion vehicles motion can be expressed as:

$$\frac{dv}{dt} = \frac{\sum F_{\text{drive}} - \sum F_{\text{res}}}{m_s}$$

v = Vehicle speed

$\sum F_{\text{drive}}$ = total drive force

$\sum F_{\text{res}}$ = total resistance force.

m_s = mass of the vehicle
 G = mass factor of the vehicle (can vary with motion)

The moment force produced by the forward shift of resultant ground force is called as rolling resistance factor.

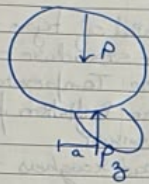
P = Normal load acting on the centre of the rolling wheel. (Unit Newton)

a = Deformation of tyre (unit m)

M = Mass of vehicle (kg)

g = Acc. due to gravity

$$T_{rr} = P a = M g a$$



To keep the wheel rolling (for) force acting on the centre of the wheel is required to balance this rolling resistance moment.

$$F_{rr} = \frac{T_{rr}}{a_{dyn}} = \frac{P a}{a_{dyn}} = P \frac{a}{a_{dyn}}$$

dynamic radius of tyre (meters)

rolling resistance coeff.

Rolling Resistance $\rightarrow F_{rr} = \frac{T_{rr}}{a_{dyn}} = \frac{M g a}{a_{dyn}} = M g f_{rr}$

$$T_{rr} = P a = M g a$$

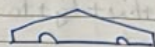
2) Aerodynamic Drag:-

→ Skin effect - Vehicle air passes over its motion but not to the air stationary. i.e. 10/22/26 that causes resistance to the vehicle

Shape of vehicle

If front area more - more drag experienced by the vehicle more surface

HP



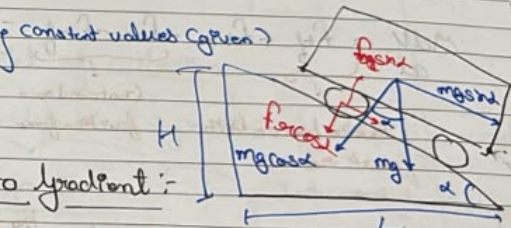
$$\text{Aerodynamic Drag} = \frac{1}{2} \rho C_d A_f v^2 = F_d$$

ρ = Density (kg/m³)

C_d = drag coefficient

A_f = vehicle front area

v = velocity



3) Opposition due to gradient:-

Gradient or grading resistance can be expressed as:-

$$F_g = M g \sin \alpha$$

4) Acceleration Resistance:- (F_a)

J = inertia of moving components

kg m²

M = mass of vehicle

$$F_a = \left(M + \sum \frac{J_{rot}}{a_{dyn}} \right) \cdot \frac{dv}{dt}$$

For going uphill we also consider normal force forces

We also consider $\alpha = H/L$ for small uphill gradient

$$F_{\text{net}} = F_{\text{traction}} - F_{\text{weight}} + F_N + F_g + F_a$$

Using second law

$$\frac{dV}{dt} = \frac{F_t - F_{\text{weight}}}{M}$$

$$M \frac{dV}{dt} = F_t + F_{\text{traction}} - F_a - F_{\text{weight}} - F_a - F_{\text{weight}}$$

tractive force both front & rear
friction force
traction force

$$W_g L = M g L \cos \alpha - (T_{\text{ax}} + T_{\text{ay}} + F_{\text{wh}} + F_{\text{hw}})$$

$$+ M g h \sin \alpha + M g \frac{dV}{dt}$$



By summing the moment of all the forces by point R the normal force on the front axle determined by

$$W_f = M g L \cos \alpha - (T_{\text{ax}} + T_{\text{ay}} + F_{\text{wh}} + F_{\text{hw}} + M g h \sin \alpha + M g \frac{dV}{dt})$$

$$M g L \cos \alpha - (T_{\text{ax}} + T_{\text{ay}} + F_{\text{wh}} + F_{\text{hw}} + M g h \sin \alpha + M g \frac{dV}{dt})$$

L

For passenger car $h_w = h_g$ and take $\cos \alpha \approx 1$

$$= \frac{M g L \cos \alpha}{L} - h_g [F_{\text{wh}} + F_{\text{tg}} + M g \frac{dV}{dt} \cos \alpha + M g \frac{dV}{dt}]$$

$$W_g = M g L \cos \alpha - \frac{h_g}{L} [F_{\text{wh}} + F_{\text{tg}} + M g \frac{dV}{dt} \cos \alpha + M g \frac{dV}{dt}]$$

$$F_{\text{fr}} = W_g \times \mu$$

Slip factor - The slip between tyre & surface, if it is muddy

$$S_{\text{af}} = \frac{C_{\text{DR}} \sigma_{\text{max}} - V}{C_{\text{DR}} \sigma_{\text{max}}}$$

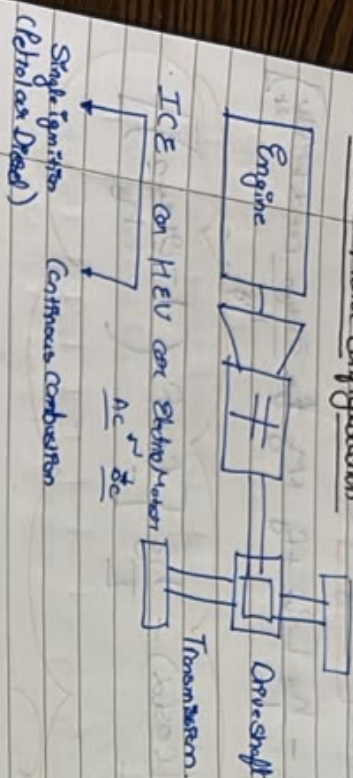
When the tractive effort of wheel exceeds, the maximum tractive effort limit imposed by adhesive capability also the tyre and the ground.

The driven wheel will spin on the ground.

Power - Torque Characteristics?

11/08/26

Power Plant Configuration



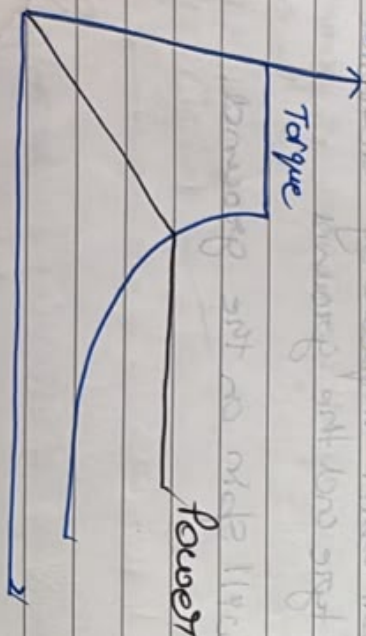
Petrol
(advantages)

- High weight ratios
- low consumption time
- Better performance.
- disadvantage
- Higher fuel consumption
- Quality of fuel required is high.

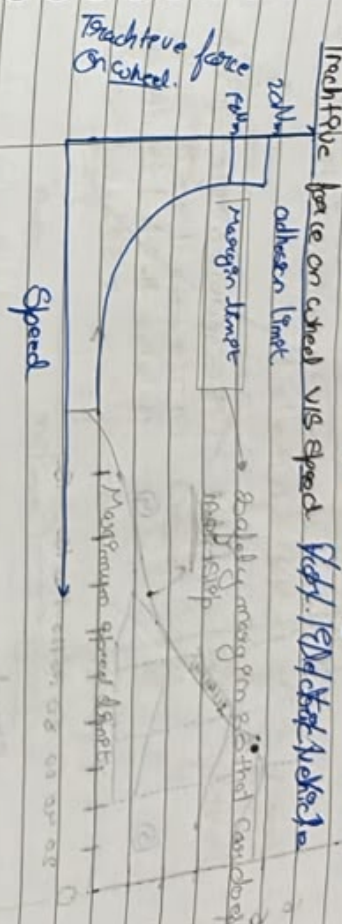
Diesel
(advantages)

- less fuel consumption.

Torque characteristics of electric motor



on vehicle app. the ideal performance characteristics of power plant is current power output over full speed range. Consequently the torque varies hyperbolically w.r.t. speed.



Cannot apply infinite tractive force of zero speed as follows. Limits thrust. They are: 1) Tractive force limit.

- 1) Adhesion force - (Friction, rolling, grading, drag - cannot attain a value).

Velocity is limited by -

- 1) Valve open and close limit for ICE
- 2) Fuel burn time.
- 3) Friction limit (Piston and clearances)

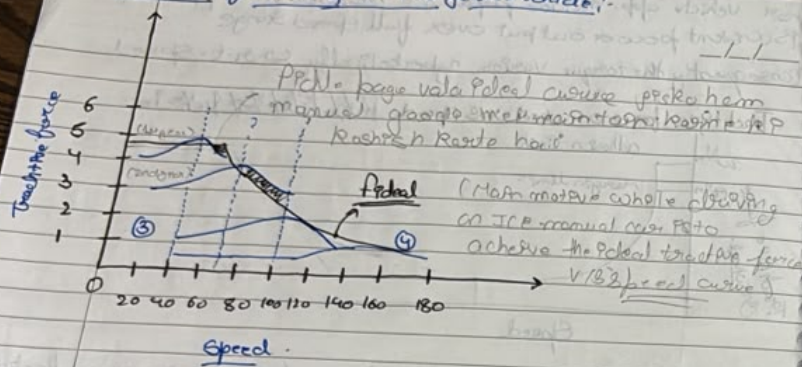
For Motor velocity is limited by

- 1) Motor heating limit
- 2) Insulation breakdown due to overheating (breakdown)
- 3) High centrifugal force at high rpm.

Common Reasons:-

- 1) Mechanical transmission system
- 2) Vibration limits
- 3) Bearing limits.

EV

[illegible]

Speed

Various Indices + Performance Indices

12) ~~Torque Increase Indices~~ - Various indices are used to characterise different types of indices.

1) Torque Increases Indices =

$$T = T_{\text{mecc}}$$

Modiumum torque

50

Torque at which power is max

$$C_n = T(P_{max})$$

2) Engine speed indices

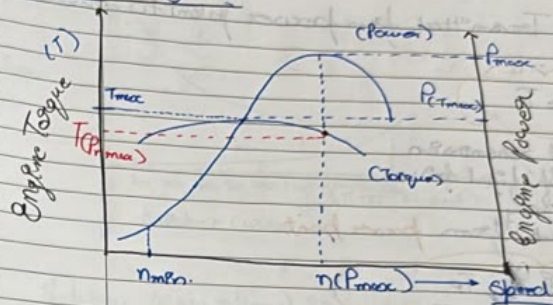
$$V = \frac{n}{n(T_{max})}$$

Speed at max torque

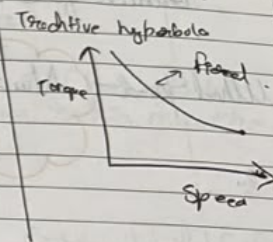
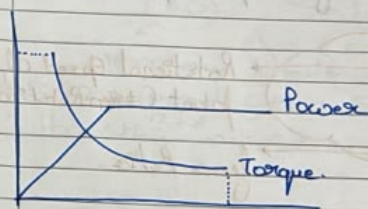
The high value of the product $\tau \times V$ - better engine power at low and med Pump Speed Speed

This enthusiasm means less frequent gear changing for IC Engine :-

12/08/20



For Electric Vehicle:-



With the maximum available engine power P_{max} and sea speed V , Ideal tractive hyperbola.

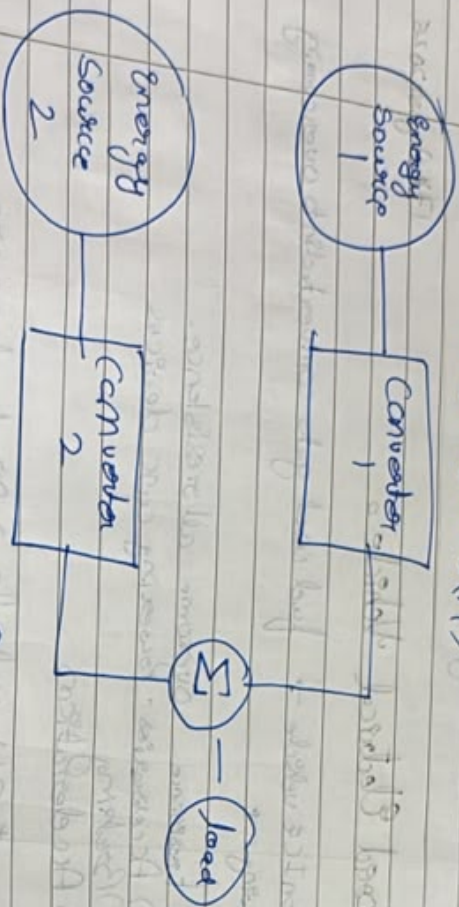
$$f_{\text{ideal}} = \frac{P_{\text{max}}}{\eta}$$

$$f_{\text{effective}} = \frac{P_{\text{max}}}{N} \times \eta_{\text{total}}$$

Efficiency of
total drivetrain

HEV $\rightarrow$ ICE + battery
 Fuel cell + battery
 cell/battery + battery
 Flywheel + battery
 battery + battery

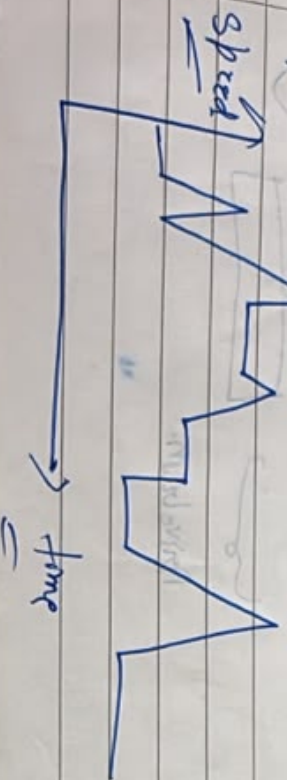
Powertrain :- for HEV
 Unidirectional (P_1)



bidirectional (P_2)

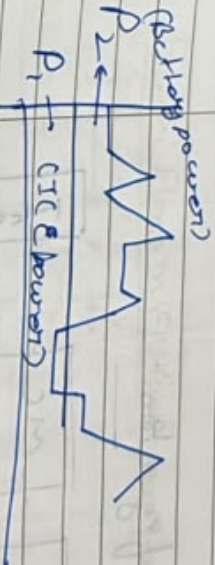
(1) $\rightarrow$ $P_1 \rightarrow$ load
 $P_2 \rightarrow$ load
 $P_1 \rightarrow P_2 \rightarrow$ load
 $P_2 \rightarrow P_1 \rightarrow$ load

Driving cycle :- (How your speed changes with time)
 Urban pattern :-



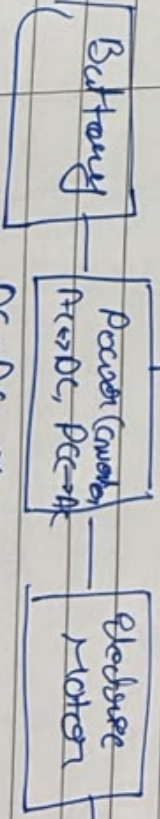
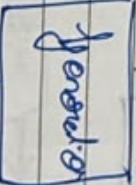
(7) $\rightarrow$ $P_1 \rightarrow$ load
 $P_2 \rightarrow$ load
 $P_1 \rightarrow P_2 \rightarrow$ load
 $P_2 \rightarrow P_1 \rightarrow$ load

Engine battery
 $\rightarrow$ $P_1 \rightarrow P_2 \rightarrow$ load



Configuration :-

(Series Configuration - dangerous)
 hence in bigger vehicles



DC-DC con
 DC to AC

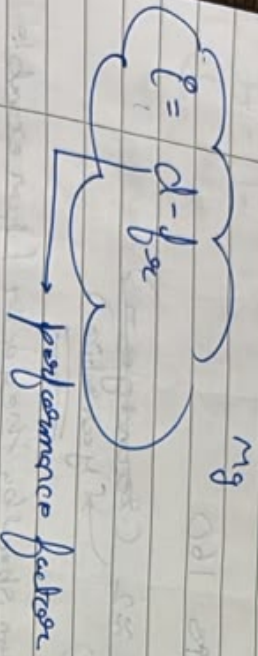
Transmission

2.) Hydroabsorption :-

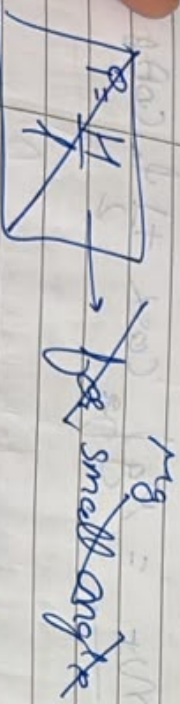
$$\frac{T_p P_{g10} m_+}{m_{g10}} = m_{g10} \cos \alpha + \frac{1}{2} \rho_a C_d A V^2 + m_{g10} \frac{g}{m_{g10}}$$

← we get small hence ignore

$$P = T_p P_{g10} m_+ / m_{g10} - \frac{1}{2} \rho_a C_d A V^2 = m_{g10} \cos \alpha$$



$$d = T_p P_{g10} m_+ / m_{g10} - \frac{1}{2} \rho_a C_d A V^2$$



$$S_{ind} = \frac{d - f_x^2}{1 + f_x^2} \sqrt{1 - d^2} \sqrt{f_x^2} \quad \left\{ \begin{array}{l} \text{for large } d \\ \text{for small } d \end{array} \right.$$

3) Acceleration Performance :-

$$a = \frac{f_x - f_w - f_g}{m_s} = \frac{g (d - f_x)}{g}$$

δ = rotational inertia factor.

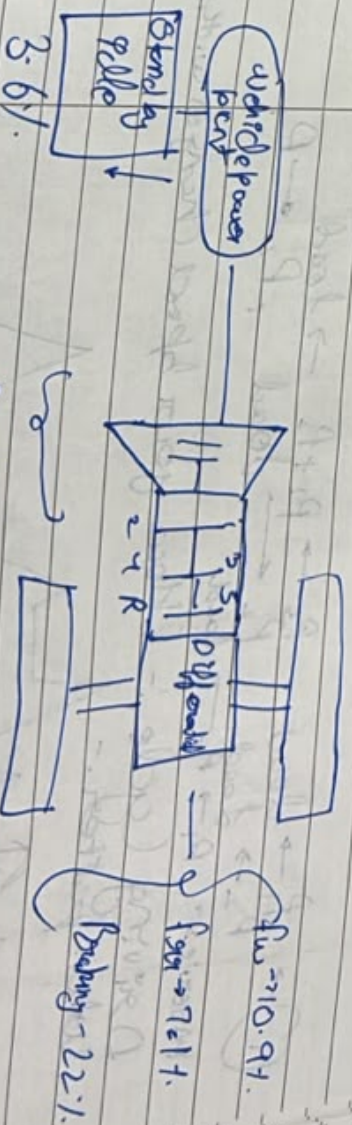
$$\delta = 1 + \frac{I_w}{m_s r_{g10}^2} + \frac{P_{g10} P_{g10} I_p}{m_s r_{g10}^2}$$

Hybrid Electrical Vehicle :-

For on ICE vehicle → fuel used gets converted into overcoming following →

- 1) Friction - overcome all resistance.
- 2) Acceleration - Powering and deceleration.
- 3) Braking
- 4) Acceleration

For on EV vehicle → Disadvantages are



13th August 2026

Mathematical Modelling:-

1.) Torque - Transmitted from power plant to driven wheel.

$$T_w = i_g i_a \eta_t T_p$$

i_g = gear ratio of transmission

i_a = gear ratio of final drive

η_t = total efficiency

T_p = torque output from power plant

$$F_t = \frac{T_w}{r_{dyn}} = \frac{i_g i_a \eta_t T_p}{r_{dyn}}$$

Tractive force in terms of torque (power plant)

Wheel Speed - $N_w = \frac{N_p}{i_g i_a}$

Rotational speed of power plant (or Rotational)

gear ratio.

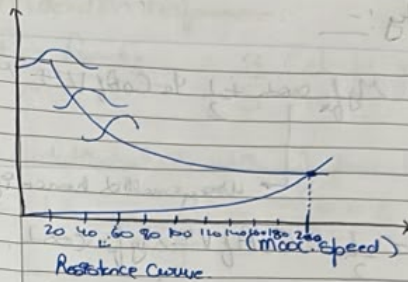
Translational Speed of wheel -

$$V = \frac{2\pi N_w r_{dyn}}{60} \text{ m/s}$$

Vehicle performance is affected by

- 1) Maximum Speed (Cruising Speed)
- 2) Gradability
- 3) Acceleration

Torque



Let say max speed is 160

At 60 Speed $\rightarrow \frac{160}{60} \approx 2.67$ (second gear)

(60) $\rightarrow$ (gear ratio)

maximum speed by the gear (for example this was second gear so gear ratio must be of 2 to attain max speed of 160.

##

$$\frac{T_p i_g i_a \eta_t}{r_{dyn}} = Mg f_r \cos \theta + \frac{1}{2} \rho_a C_d A v^2$$

Rolling Resistance Drag

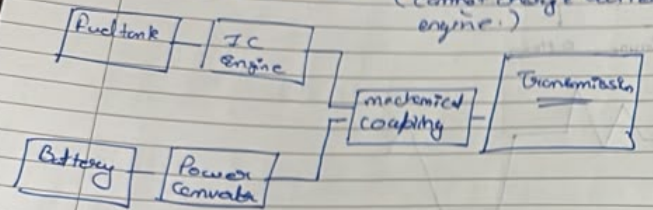
$$V_{max} = \frac{2\pi N_{max} r_{dyn}}{60 i_{gmax} i_a}$$

gear ratio max speed in 1st gear

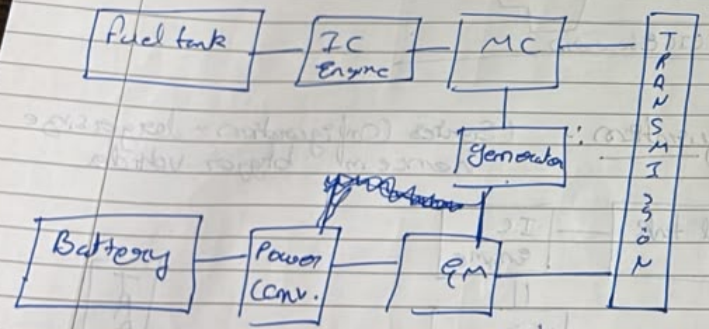
Gen → Charges battery and provides wheel torque

Parallel Configuration:-

Used in battery vehicles (cannot charge battery from engine.)



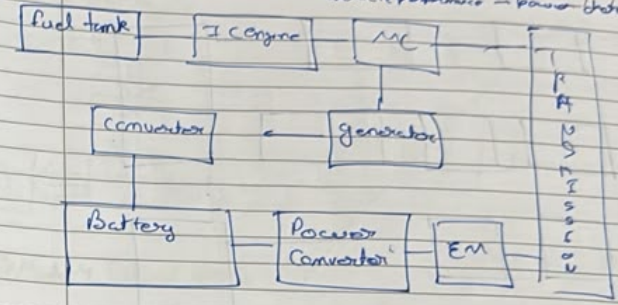
Series-Parallel Configuration:-



$P_1 \rightarrow P_2 \rightarrow \text{load}$
 $P_2 \leftarrow P_1$

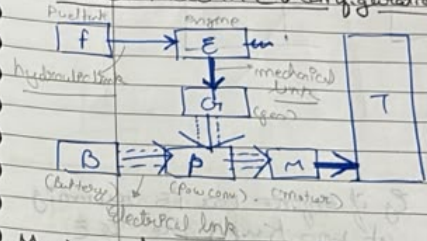
Complex design:-

Social - Impact, comparison
 Env. - Numerical
 Eco. - price
 Vehicle performance - power distribution



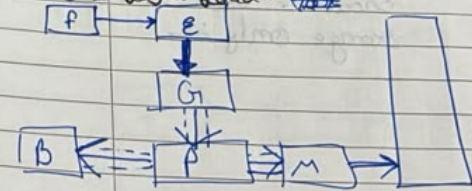
Power flow in EV configuration:-

(Power flow control)

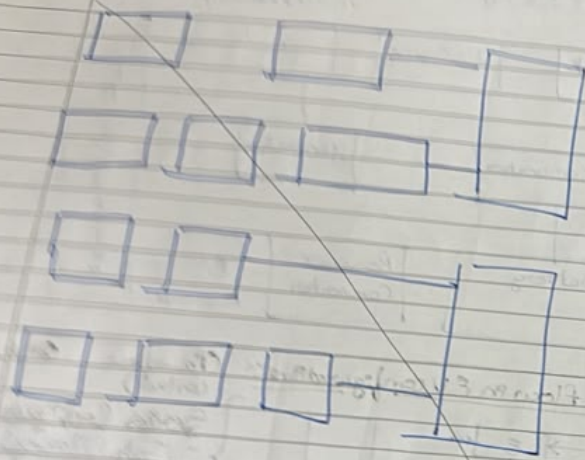


Series Configuration
 - Model Normal acceleration
 $(P_1 + P_2 \rightarrow \text{load})$

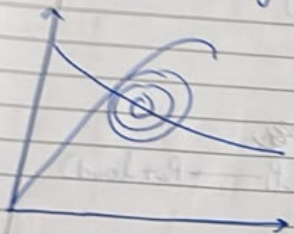
Mode-2 Light load.



$(P_1 \rightarrow P_2 + \text{load})$



1) ICE operation-region.

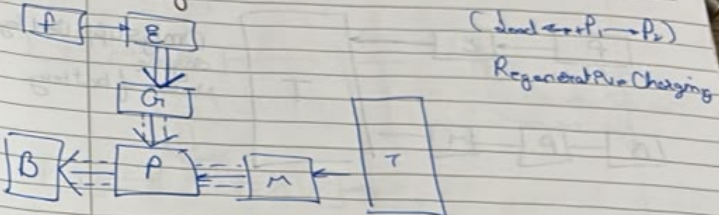


Eg if engine car is running at 200 kmph the ICE should be used at a certain range only i.e.

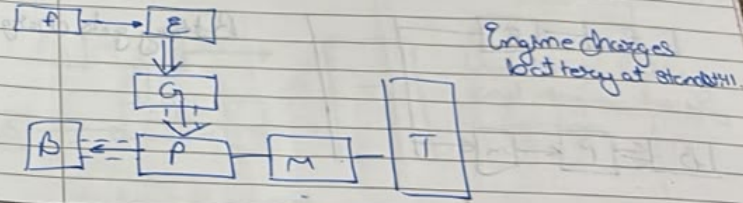
2) ICE operating line

3) Battery operating - Not charge more than a certain level and not a charge more than a certain level

Mode-3 Braking:-

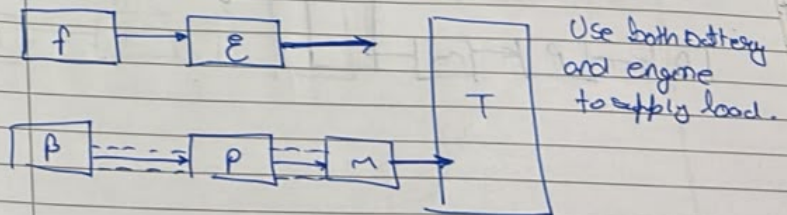


Mode-4 Standstill:-

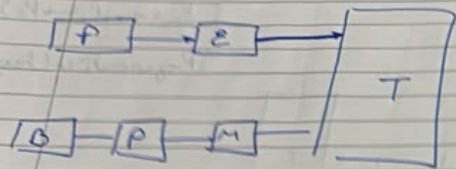


Parallel Configuration:-

Mode-1 Start up / full throttle / acceleration.

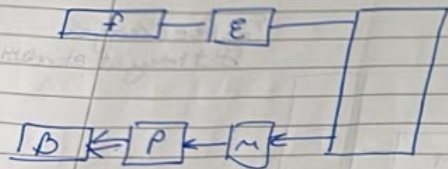


Mode-2 (Normal Acceleration)



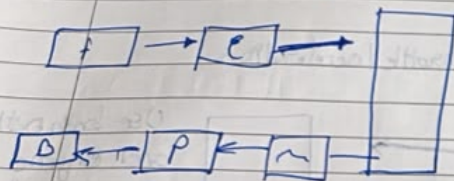
Only supplied by engine
No use of battery

Mode-3 Braking



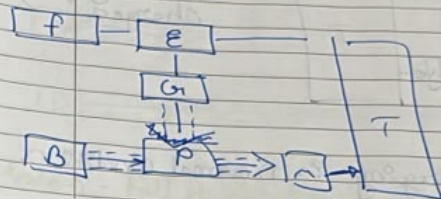
Battery gets charged

Mode-4 Light Loading



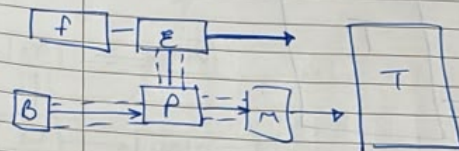
Battery gets charged with load.

Series Parallel Mode-1 Start-up



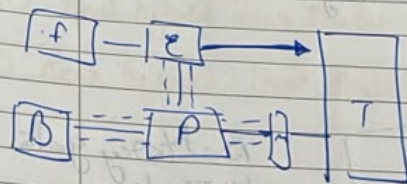
Only uses battery to start. Engine remains off

Mode-2 Full acceleration



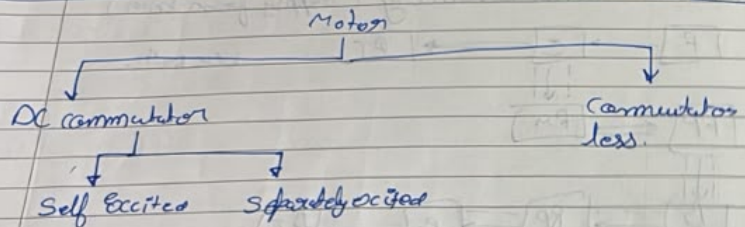
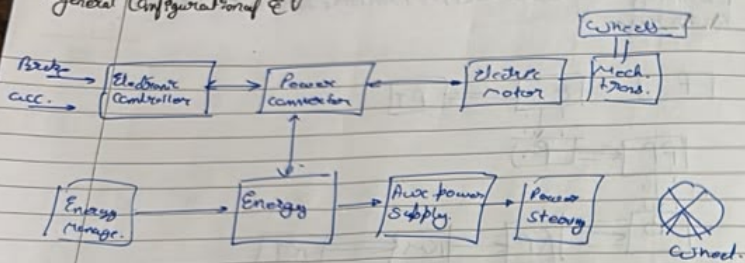
Uses both Battery and engine to supply power.

Mode-3 Normal acceleration



Battery at standstill only uses ICE

General Configuration of EV



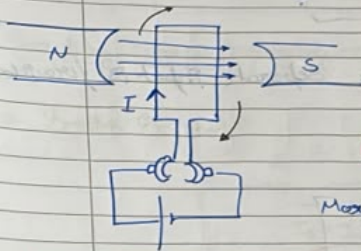
DC Motor Drive :-

- High speed regulated Rm
- Adjustable speed
- frequent starting
- Braking & regenerative

Problems: Low specific power
(more weight less power)

25/09/2022

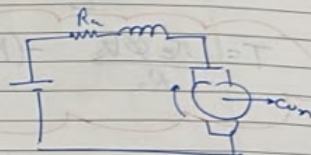
(FLH sub-egon)
(FRH sub-mot)



Torque produce :-

$$T = BIL \cos \theta$$

More coils (wdg.) more torque.



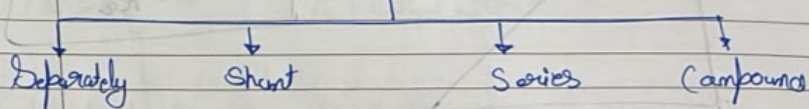
$$V_a = I_a R_a + K_e \phi \omega_m$$

$$E = K_e \phi \omega_m$$

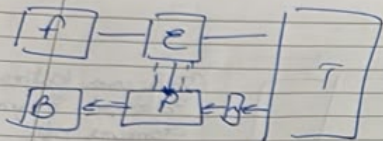
$$T = K_t \phi I_a$$

If permanent magnet is not used and wound electromagnets are used.

Wound Motor

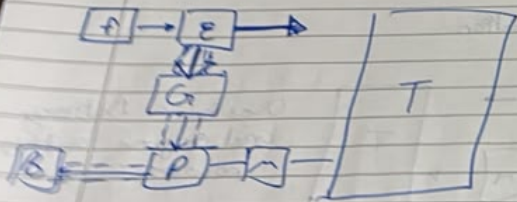


Mode-4 Braking



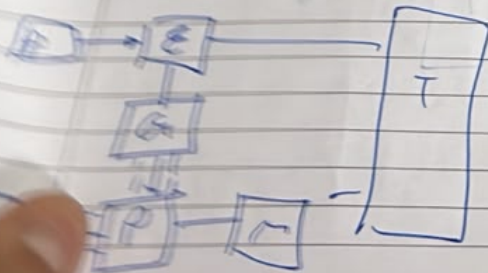
(Battery gets charged)

Mode-5 Battery charging in normal condition



use power and PS more then charge battery top occur and when it becomes less use ICE to both side and charge battery

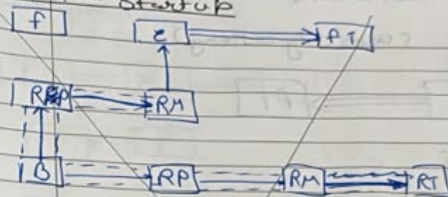
Mode-6 Standstill



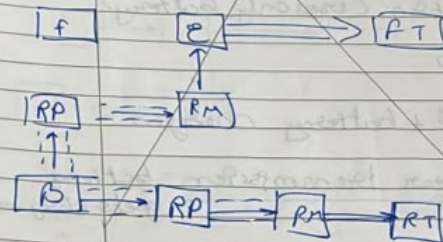
Battery gets charged.

Complex System 2 (Front and Rear Transmissions)

Mode-1 Startup



Mode-2 - Full Acceleration (uses only battery)

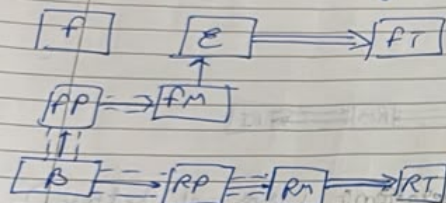


These are different modes to operate complex hybrid system

- ① Front hybrid Rear Electric $\rightarrow$ ICE heavy
- ② Rear hybrid - front Electric $\rightarrow$ battery heavy.

→ Front hybrid power electric

Mode-1 → Startup (uses only battery)

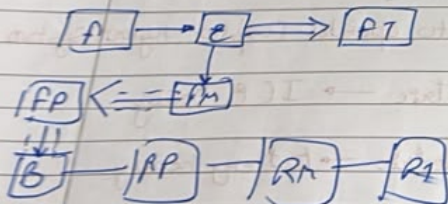


Mode-2 - Maximal Acceleration (uses only battery)

(Same as startup)

Mode-3 Vehicle propels + battery charges

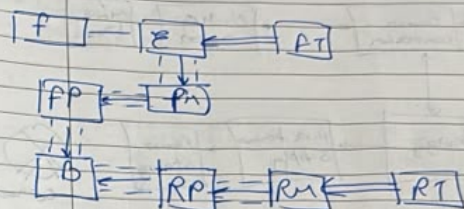
→ No use of gear transmission battery gets charged



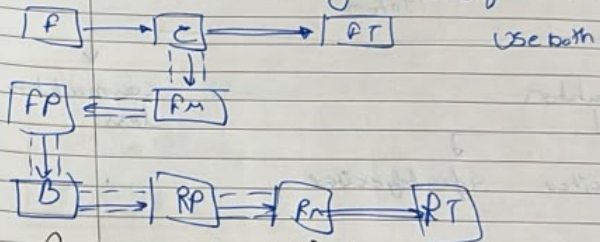
Mode-4

Full Acceleration — Use only battery No use of engine

Braking — Uses both transmission to charge battery



Mode-5 Axial Balancing (supply from both)



Rear hybrid power electric:-

In Mode-1 - Startup (Battery to gear transmission)

Mode-2 — ~~use both~~ fuel give power to both FT, RT and battery also provides to FT.

Mode-3 Vehicle propels + Bat. chrg:-

f → FT

f → Battery

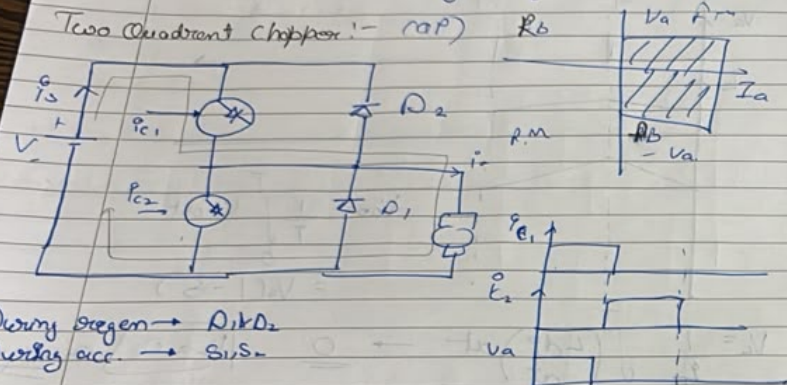
Mode-4 - Light load. → f → FT

Mode-5 Braking → FT → B
RT → B

Mode-6 Axial Balancing f → FT, f → B

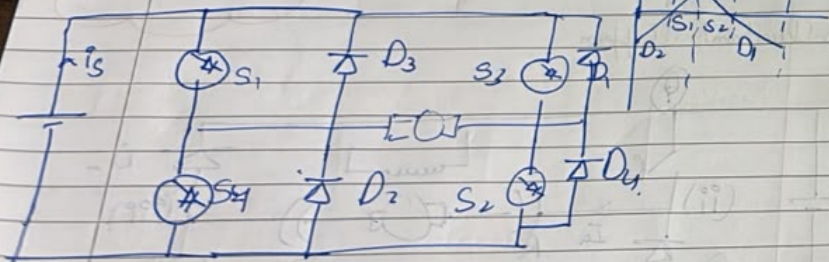
Current will flow from (1) (17) (17), (17) when switch is closed
 When switch is open → Current directed through armature
 generates same but voltage charged hence

Two Quadrant Chopper:- (MP)



During regen → $D1, D2$
 During acc. → $S1, S2$

Class E four Quadrant



Induction Motor:-

→ Small S_{go}

→ low cost

→ high efficiency & flexibility

Wound Rotor - speed control (external resistance) — Brushless Comm. off
 Squirrel Cage — this is used more often due to lack of brushes

↳ cost effective — controlled using VV-VF (variable voltage-variable frequency control)

FOC - field oriented control (more used)

Complexity due to non-linearity of induction motor, VVVF method is not used and FOC is more prominent.

Principles of Induction Motor:-

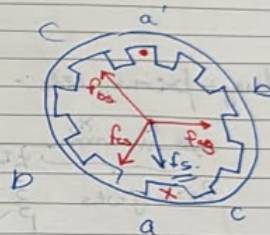
$$f_{as} = I_m \sin \omega t \rightarrow f_{as} = f_m \sin \omega t$$

$$f_{bs} = I_m \sin (\omega t + 120^\circ)$$

$$f_{cs} = I_m \sin (\omega t - 120^\circ)$$

$$f_s = \frac{3}{2} f_m e^{j(\omega t - 90^\circ)}$$

(phasor)



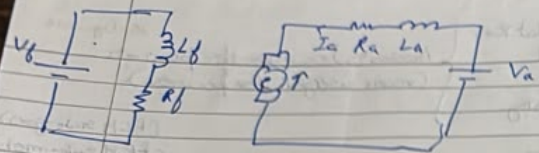
Resulting Magnetic field:-

when $\omega t = 0^\circ$ $f_{as} = 0$, $f_{bs} = -f_m/2$, $f_{cs} = f_m/2$

at $\omega t = 90^\circ$ $f_{as} = f_m$, $f_{bs} = -f_m/2$, $f_{cs} = -f_m/2$

$$f_s = 0 \text{ f.s.}$$

$$\omega_m = \frac{p}{2} \omega_e$$



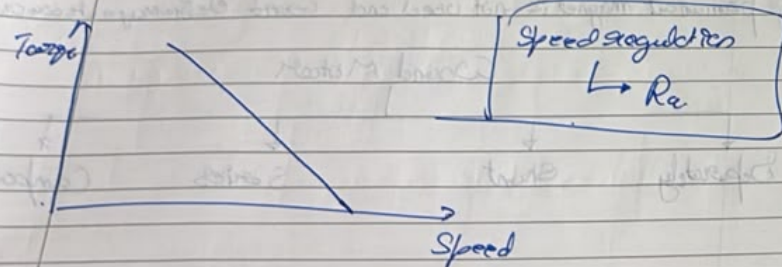
In separately excited DC motor separate supply is provided to field and armature.

$V_f = \text{constant}$, $\phi = \text{constant}$

$$T = K_e \phi \left[\frac{V_a - K_e \phi \omega_m}{R} \right]$$

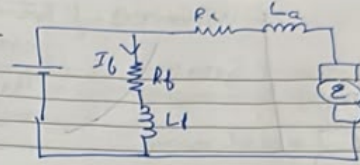
$$T = \frac{K_e \phi V_a}{R_a} - \frac{(K_e \phi)^2 \omega_m}{R_a}$$

It is a linear equation with only changing factor ω_m & T & ω_m



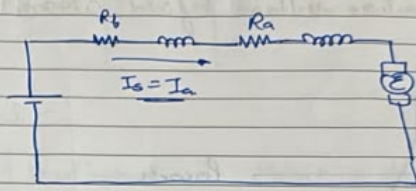
Application - If we require sep. excited DC motor for good speed regulation & non adjustable speed.

Shunt:-



Everything same as separately excited.

Series Excited:-

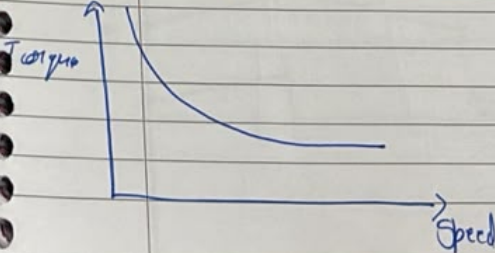


$$\phi = f(I_a)$$

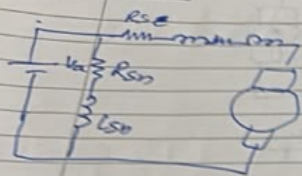
$\uparrow T = K_e \phi I_a \uparrow$ (This provides high torque for high speed)

$$T = \frac{K_f K_e V_a^2}{(R_a + K_f K_e \omega_m)^2} \quad (\text{parabolic})$$

Its disadvantage is when load is less speed is very high.

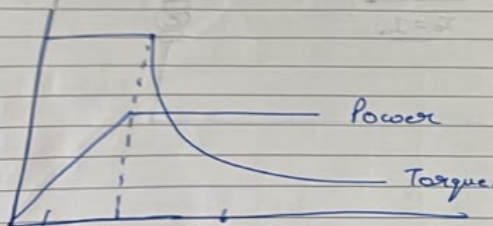


Compound :-



(See long and short shunt).

Combined armature voltage & field current control.



Armature control
 $V_a = I_a R_a + E$
 $E = K_e \phi \omega_m$
 $T = K_e \phi I_a$

Field control.

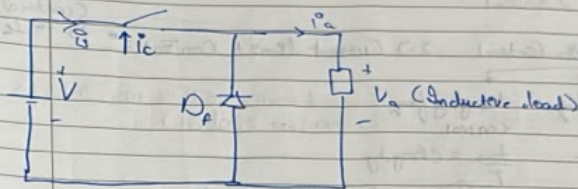
For constant torque we use
 armature voltage control using
 PE devices (choppers)

For field control we get a constant power output

Chopper Control for DC Motors:-

- Advantages are:-
- High efficiency
- Light weight
- Quick response

- Possibility of control
- Chopper $\rightarrow$ high frequency operation $\leftarrow$ high frequency ripple
- Lower losses

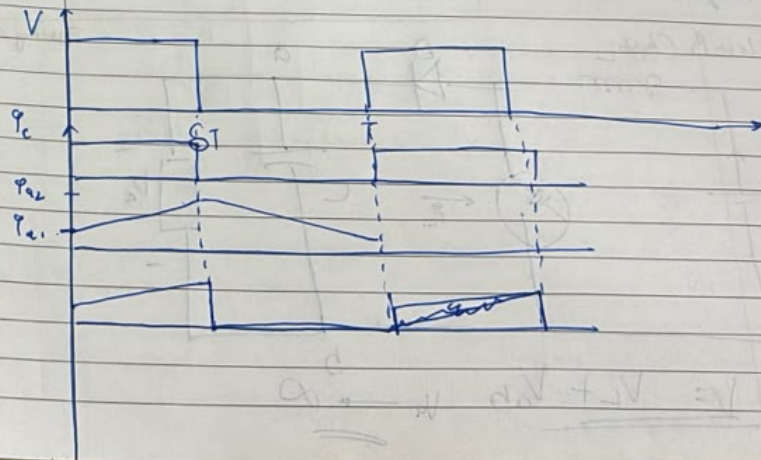


Control Signal:-

Transistor - Base Current
 GTO - Gate Current
 Thyristor - Gate Current

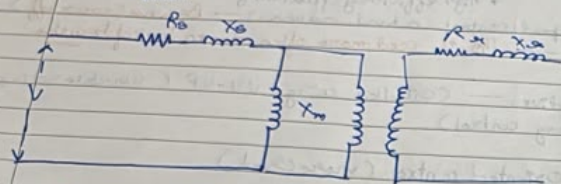
MOSFET - Gate to Source Voltage

$\delta = \text{duty}$



Steady State Performance

Eq. CN - Synchronous L.R.



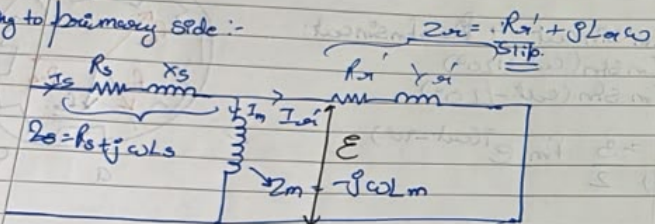
Equations:

$$a_{Ti} = \frac{n_2}{n_r}$$

For squirrel cage motor

$$n_{sc} = 1$$

Referring to primary side:-



$$I_o = \frac{V}{Z}, \quad I_a' = \frac{Z_m}{Z_m + Z_r'} I_s$$

$$R_a' = a_{Ti}^2 R_r$$

$$X_a' = a_{Ti}^2 X_r$$

$$P_{\text{power}} = 3 I_a'^2 \frac{R_a'}{s}$$

$$T = \frac{P_{\text{mech}}}{\omega}$$

Revolving Magnet field rotates at Synchronous Speed.

$\omega_m s = \text{Synchronous Speed}$

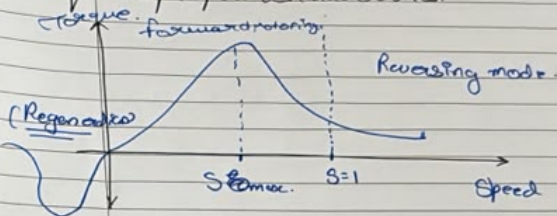
$\omega_{ms} = \text{Rotor Speed}$

$\omega_{sl} = \text{Slip Speed}$

$$\omega_{sl} = \omega_m s - \omega_{ms}$$

$$\omega_{sl} = \omega_m s - \omega_{ms} = 0$$

Torque Slip/Speed Characteristics:



$$\omega_m = \frac{2}{p} \omega (1-s)$$

When $s = 1$, $\omega_m = 0$

At $s > 1$, $\omega_m = -ve$ we enter reversing of motor.

When rotor speed > Synchronous speed. → Regenerative mode.

One of the main advantages of choppers is that they can provide a constant voltage to the load even when the input voltage is not constant. This is achieved by controlling the duty cycle of the chopper.

Type of Control:-

1) Time Ratio Control 2) Current Limit Control $\rightarrow \delta \rightarrow I_L$ (control)

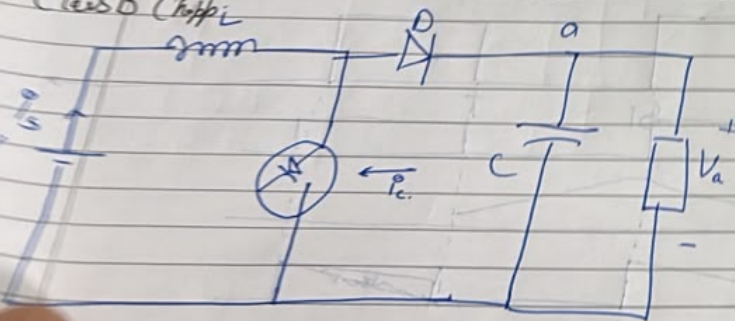
Control by Time Ratio Control (where δ is the duty cycle) $\rightarrow$ (where δ is the duty cycle)

$\frac{t_{on}}{T} = \text{constant}$ $\frac{t_{on}}{T} = \text{charging}$ (where δ is the duty cycle)

Charging a capacitor t_{on} is the time when the voltage is high. t_{off} is the time when the voltage is low. T is the total time.

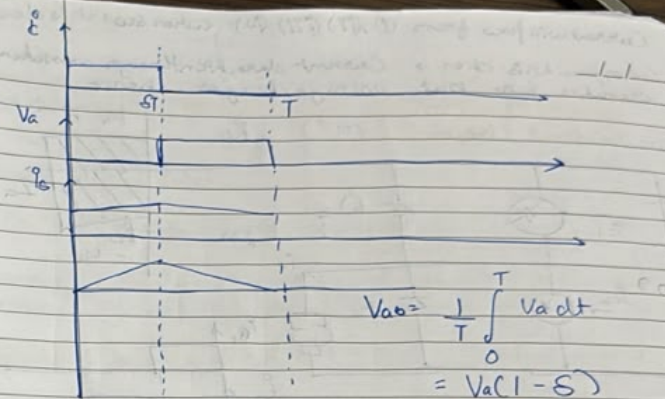
This is a first quadrant chopper control having +ve speed, torque, current and voltage.

Class B Chopper



31st Aug 21

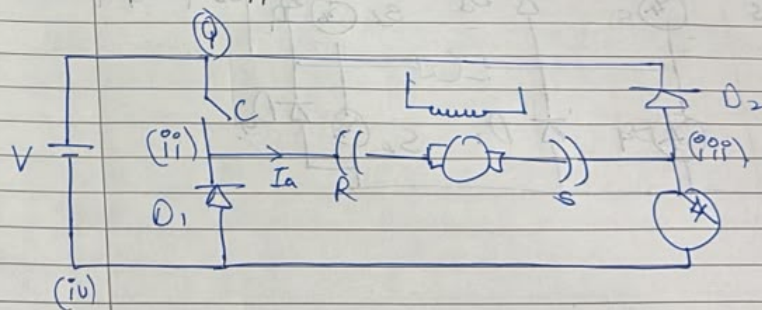
$$V_a = V_L + V_{ab}$$



$$V_L = \frac{1}{T} \int_0^T (L \frac{di}{dt}) dt \rightarrow 0 \quad \boxed{V_L = 0} \text{ average}$$

Advantages - (i) Ripples are less in source voltage. $V_a = \frac{V}{(1-\delta)}$

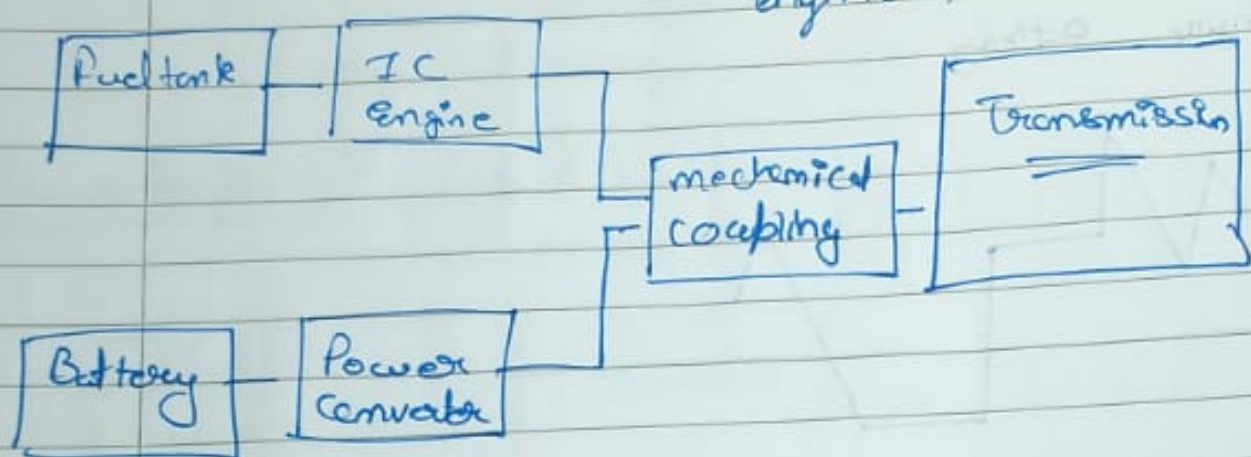
Two Quadrant Chopper



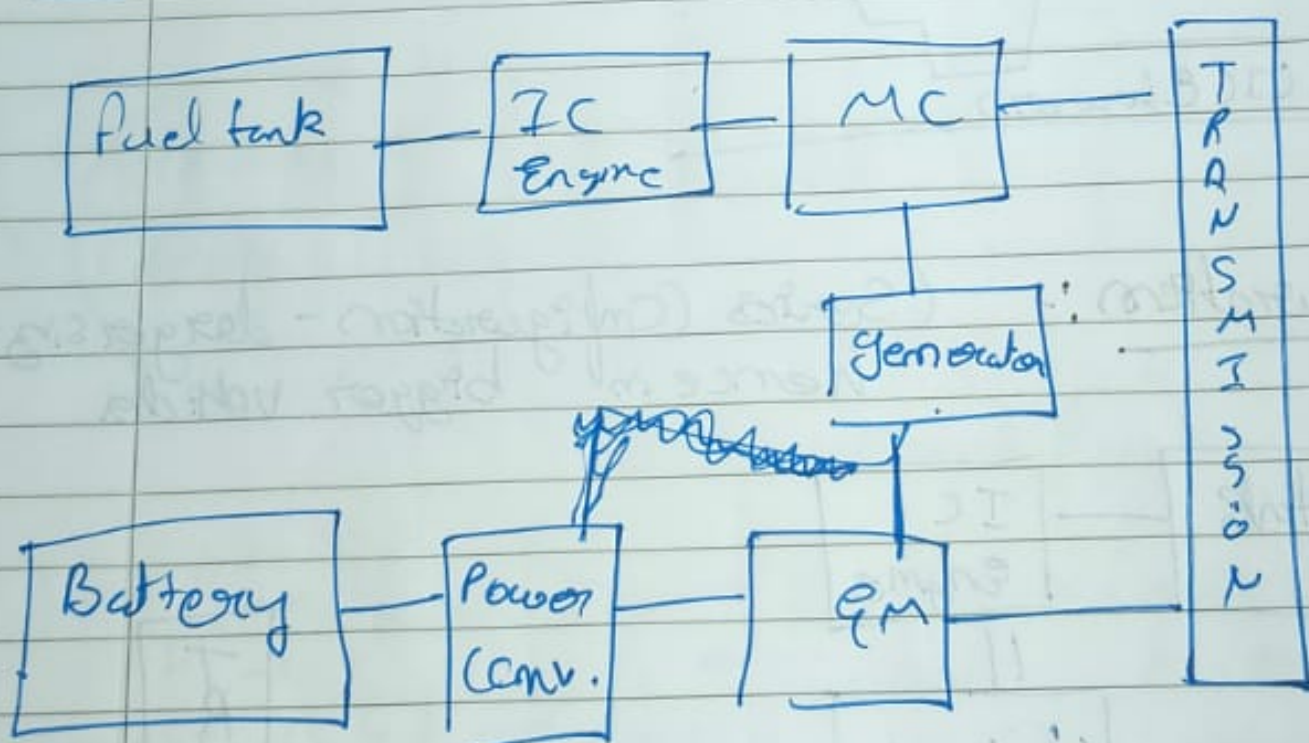
gem $\rightarrow$ Charges battery and drives wheels too

Parallel Configuration:-

Used in passenger vehicle
(cannot charge battery from engine.)



Series-Parallel Configuration:-

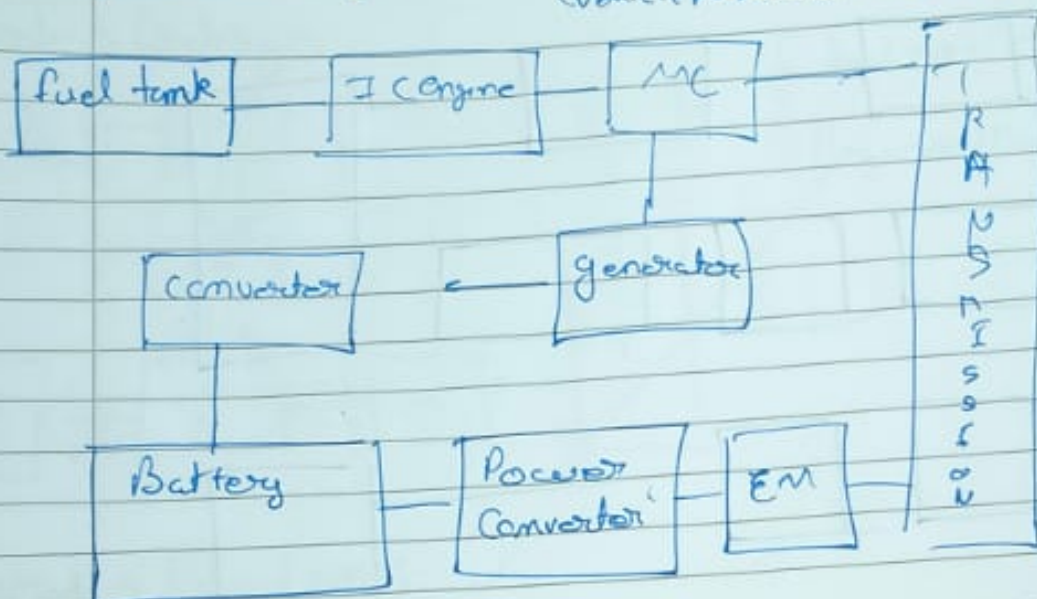


$P_1 \rightarrow P_2 \rightarrow \text{load}$

$P_2 \leftarrow P_1$

Complex design:-

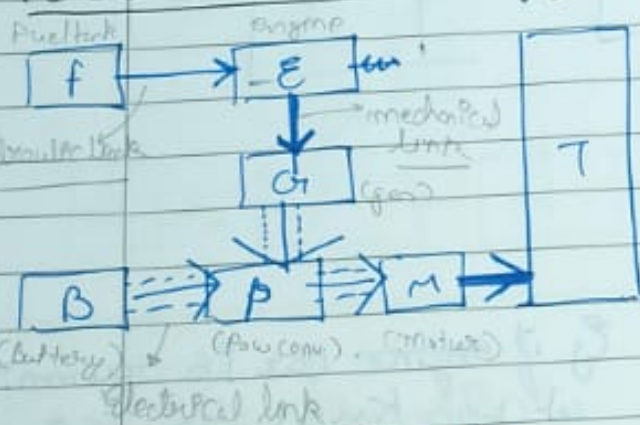
eg { Social impact, computational, eco-price, numerical, vehicle performance - power distribution



Power flow in EV configurations:-

(Power flow control)

18th Aug 2024

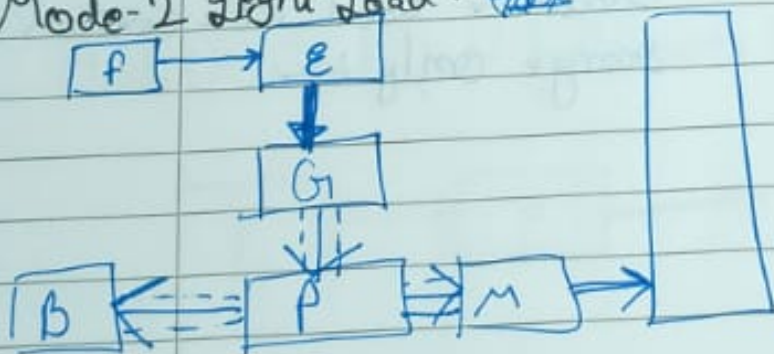


Series Configuration

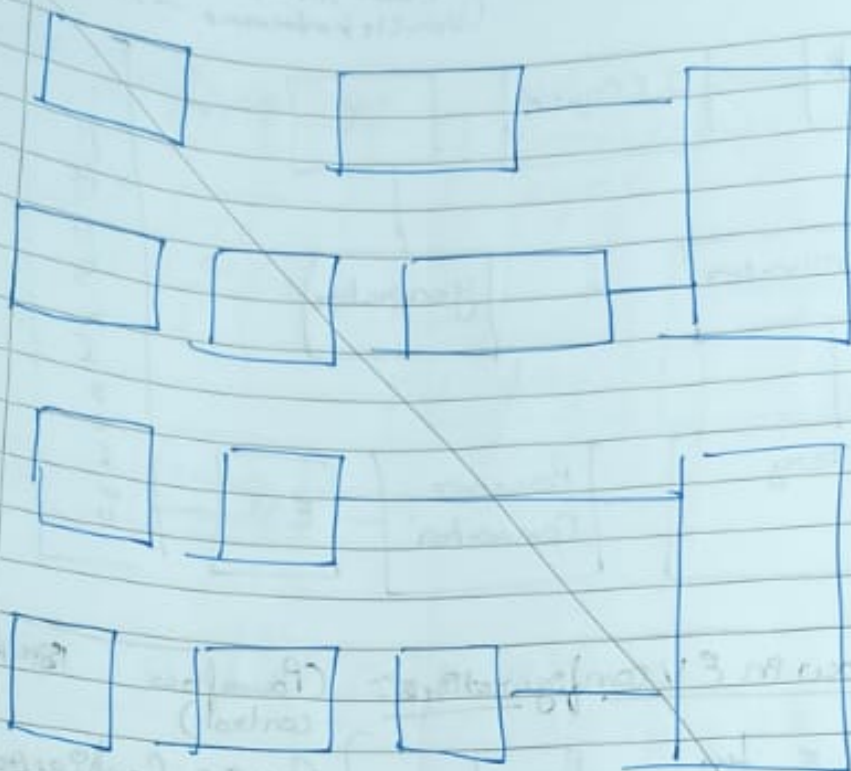
- Mode 1 Normal acceleration

($P_1 + P_2 \rightarrow \text{load}$)

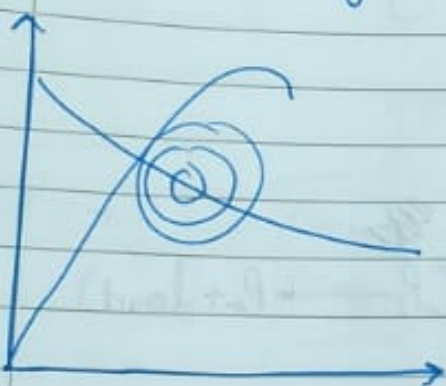
Mode-2 Light Load



($P_1 \rightarrow P_2 + \text{load}$)



1) ICE operation-region.

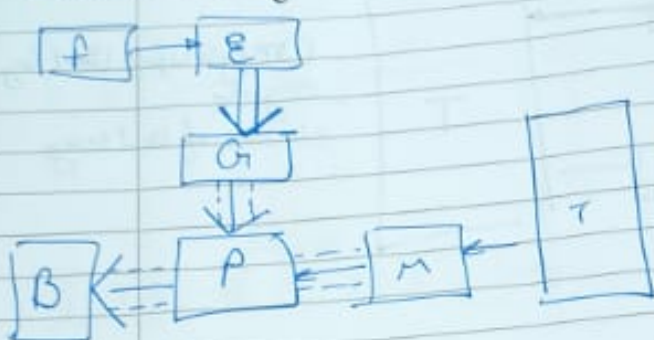


Eg if engine car is running at 200 kmph the ICE should be used at a certain range only i.e.

2) ICE operating line

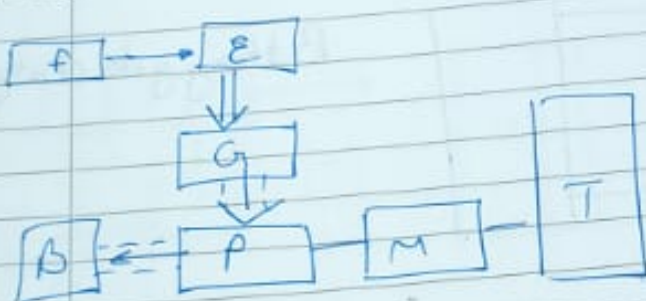
3) Battery operating - Not charge more than a certain level and not discharge more than a certain level.

Mode-3 Braking:-



(Load $\leftarrow P_1 \rightarrow P_2$)
Regenerative Charging

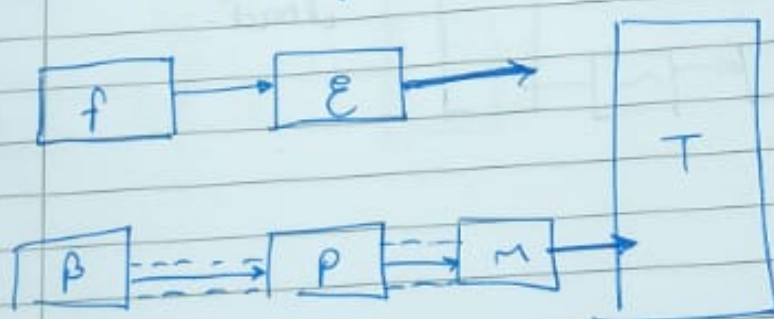
Mode-4 Standstill:-



Engine charges
battery at standstill.

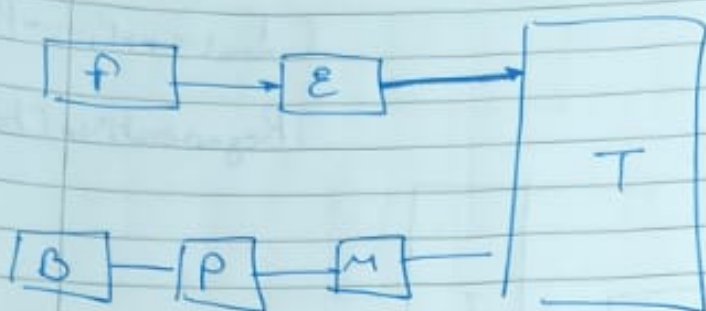
Parallel Configuration:-

Mode-1 Start up / full throttle / acceleration.



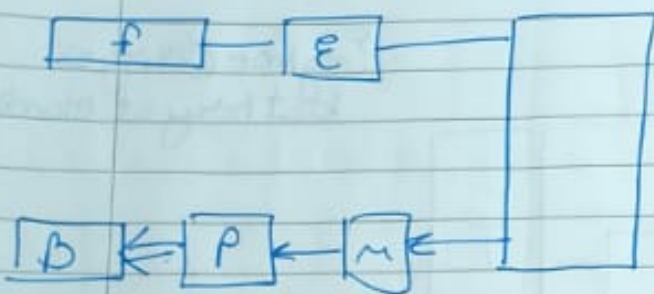
Use both battery
and engine
to supply load.

Mode-2 (Normal Acceleration)



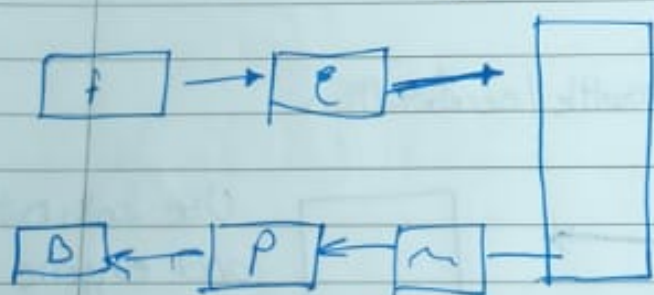
Only supplied by engine
No use of battery

Mode-3 Braking



Battery gets charged.

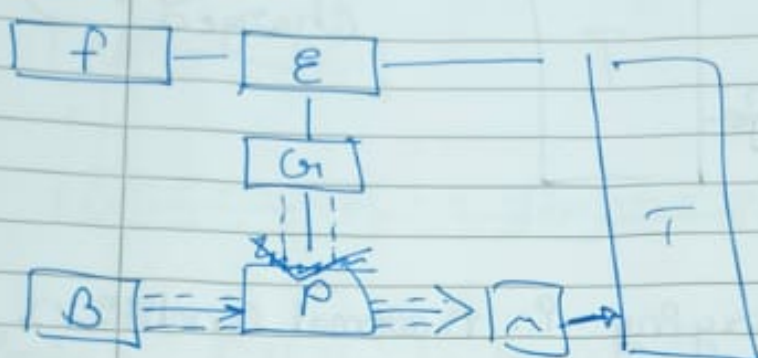
Mode-4 Light Loading



Battery gets charged with load.

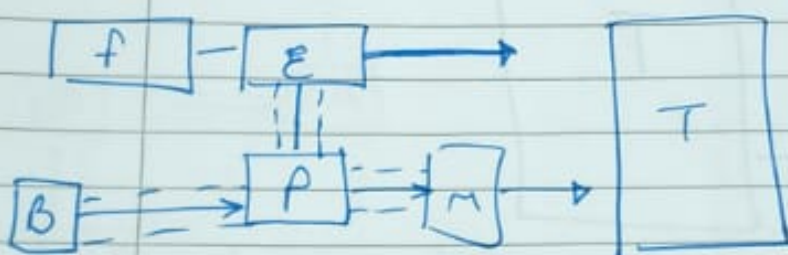
Series Parallel

Mode-1 Start-up



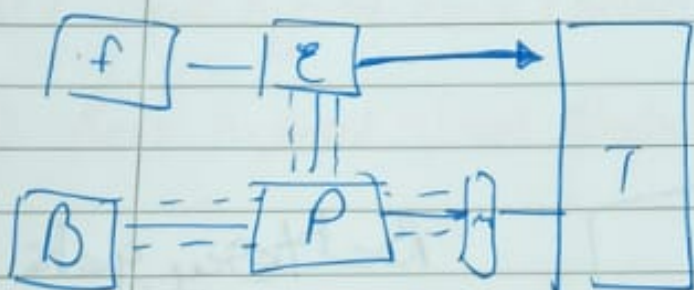
Only uses battery to start. Engine remains off.

Mode-2 Full acceleration.



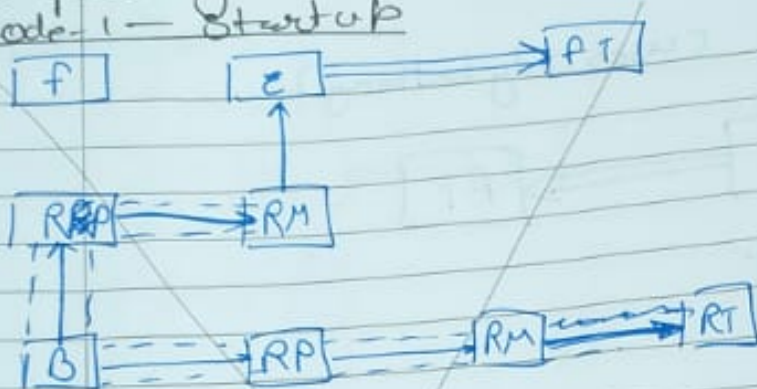
Uses both Battery and engine to supply power.

Mode-3 Normal acceleration

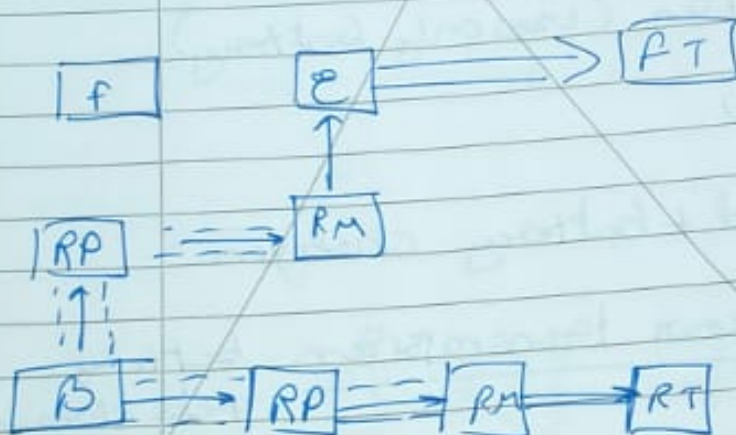


Battery at standstill only uses ICE

Complex System :- (Front and Rear Transmission)
 Mode-1 - Startup



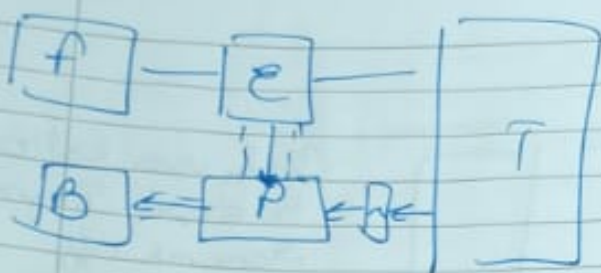
Mode-2 - Full Acceleration: (uses only battery)



There are different modes to operate complex hybrid system

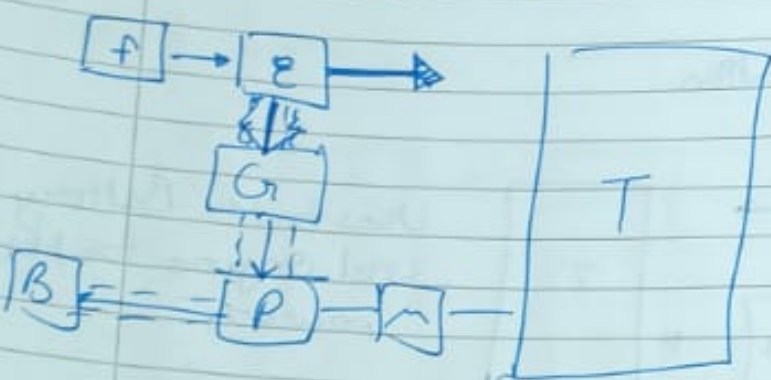
- ① Front hybrid Rear Electric $\rightarrow$ ICE heavy
- ② Rear hybrid - Front Electric $\rightarrow$ Battery heavy.

Mode-4 Braking



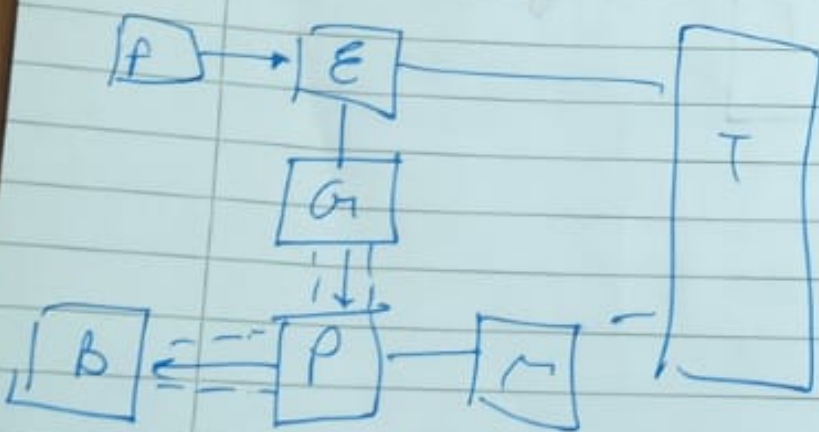
(Battery gets charged)

Mode-5 Battery charging in normal condition.



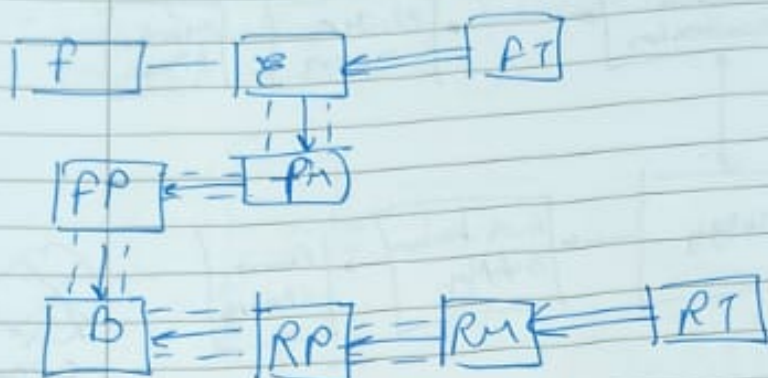
If power demand is more then ^{use} charge battery to power load when it becomes less use ICE to both drive load and charge battery

Mode-6 Standstill :-

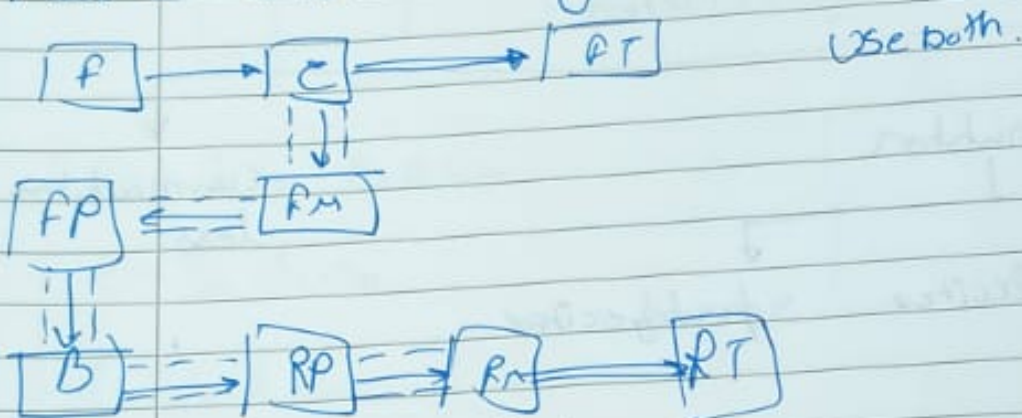


Battery gets charged.

Mode-4 Braking — Uses both transmissions to charge battery on



Mode-5 Axial Balancing (supply from both)



Recharge Hybrid Plant Electro:-

In Mode-1 - Startup (Battery to create transmission)
 Mode-2 - ~~use both~~ fuel give power to both FT, RT and battery also provides to FT.

Mode-3 - Vehicle propels + Bat. charging:-
 $f \rightarrow fT$
 $f \rightarrow \text{Battery}$

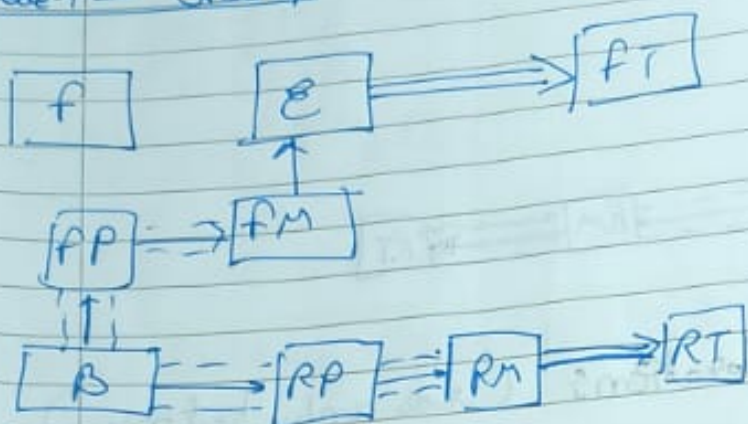
Mode-4 - Light load. $\rightarrow f \rightarrow fT$

Mode-5 - Braking $\rightarrow fT \rightarrow B$
 $RT \rightarrow B$

Mode-6 - Axial Balance $f \rightarrow fT, f \rightarrow B$

→ Front hybrid series electric

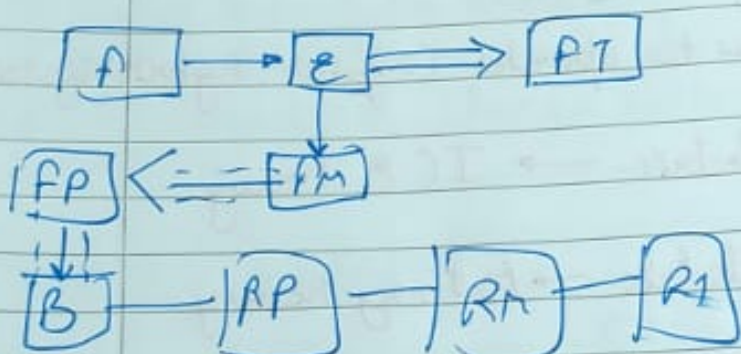
Mode-1 - Startup (uses only battery)



Mode-2 - Normal Acceleration (uses only battery)
(Same as startup)

Mode-3 Vehicle propel + battery charger

No use of gear transmission battery gets charged



Mode-4

Full Acceleration — use only battery No use of engine

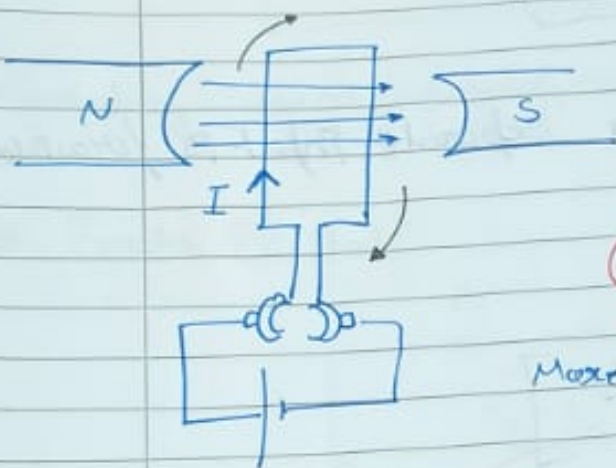
DC Motor Drive :-

25/09/2026

- High Speed regulation
- Adjustable speed
- frequent starting
- Braking & reversing

Problem is low specific power
(more weight less power)

(FLH rule-gon)
(FRH rule-mot.)



Torque produce :-

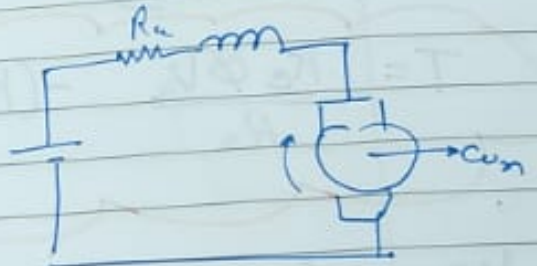
$$T = BIL \cos \theta$$

More coils (wdg.) more torque.

$$V_a = I_a R_a + K_e \phi \omega_m$$

$$E = K_e \phi \omega_m$$

$$T = K_e \phi I$$



If permanent magnet is not used and wound electromagnets are used.

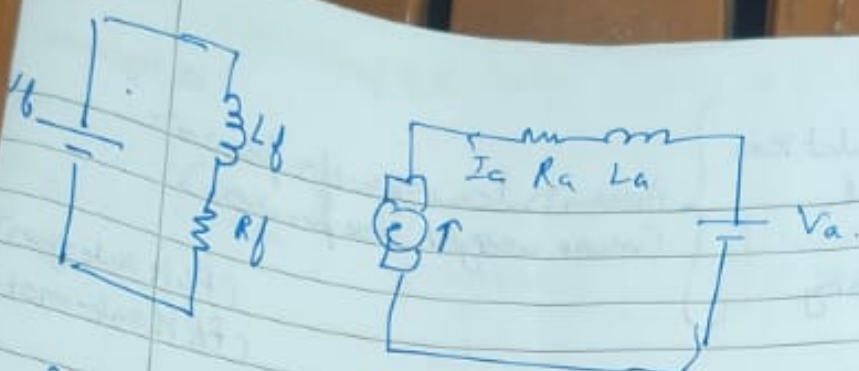
Wound Motor

Separately

Shunt

Series

Compound



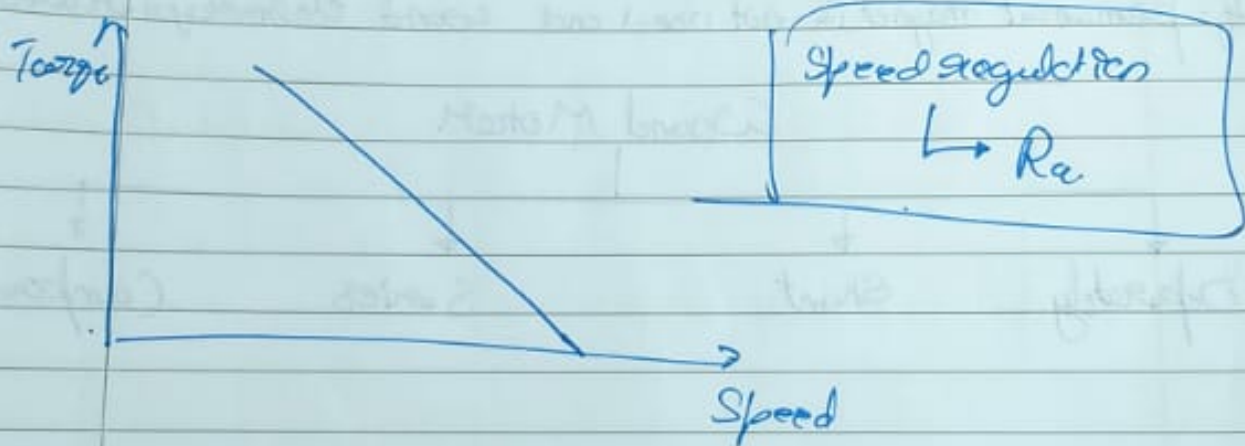
In Separately excited DC motor separate input is provided to field and armature.

$V_f = \text{constant}$, $\phi = \text{constant}$

$$T = K_e \phi \left[\frac{V_a - K_e \phi \omega_m}{R} \right]$$

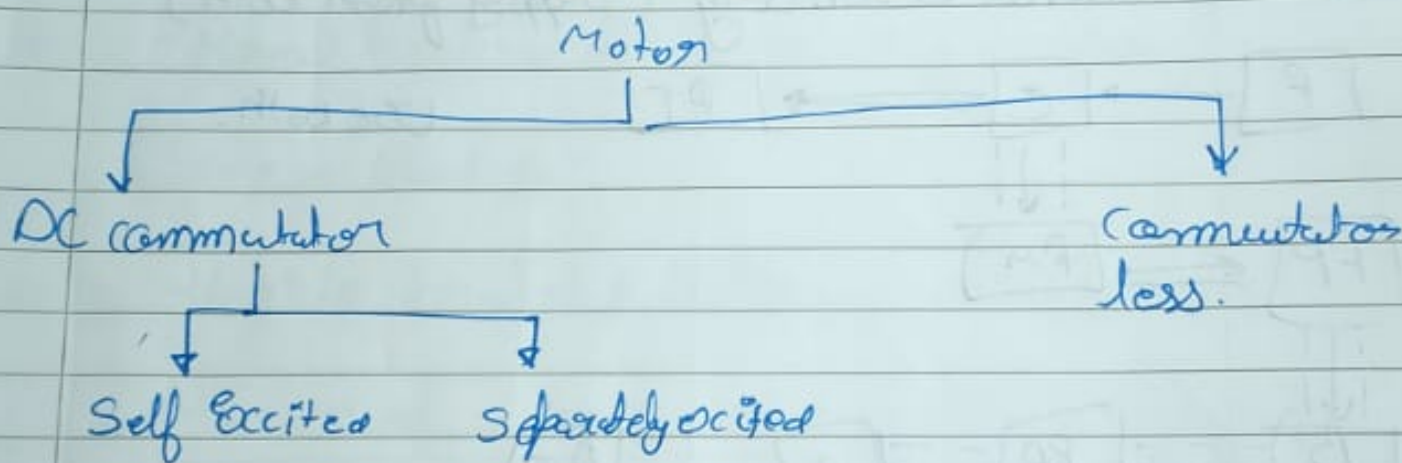
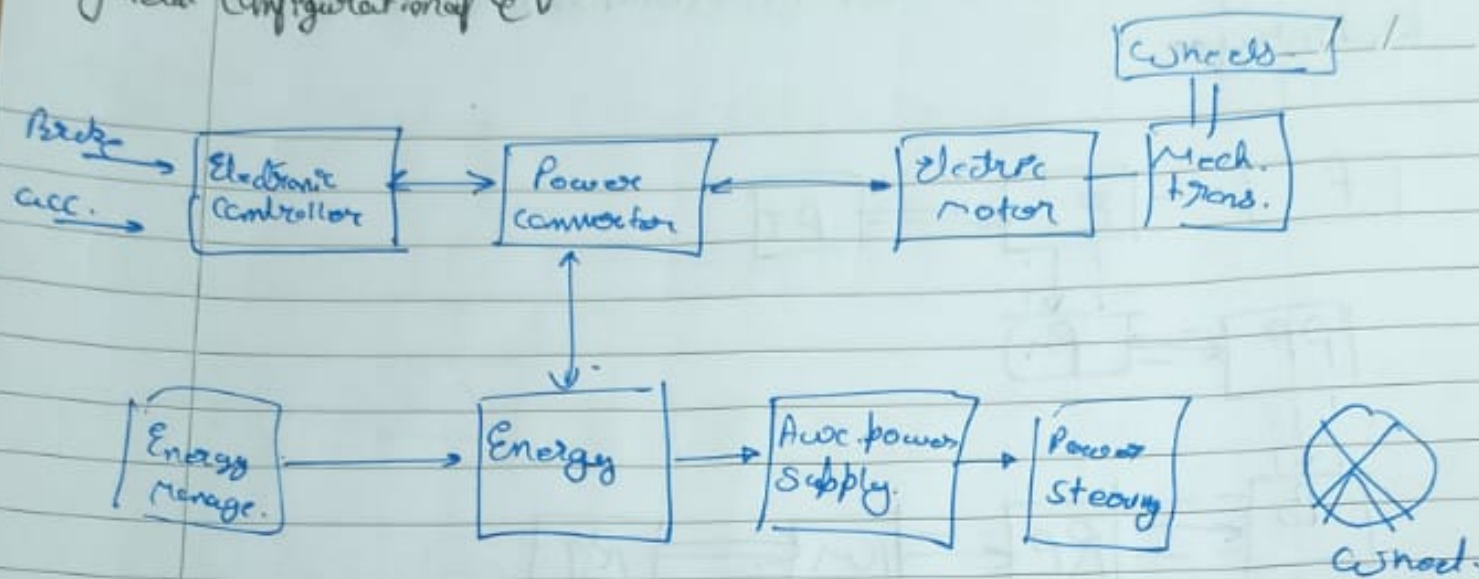
$$T = \frac{K_e \phi V_a}{R_a} - \frac{(K_e \phi)^2 \omega_m}{R_a}$$

It is a linear equation as only changing factors are T & ω_m

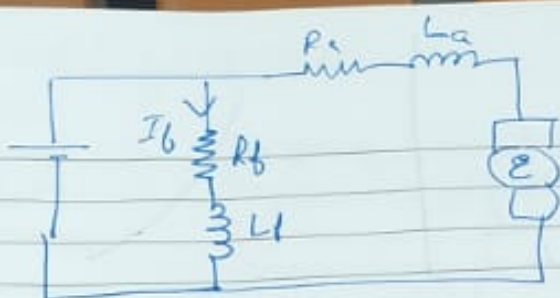


Application - If we require sep. excited DC motor for good speed regulation for proper adjustable speed.

General Configuration of EV

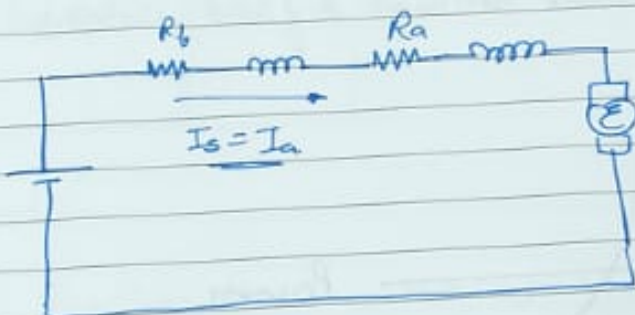


Shunt:-



Everything same as separately excited.

Series Excited:-

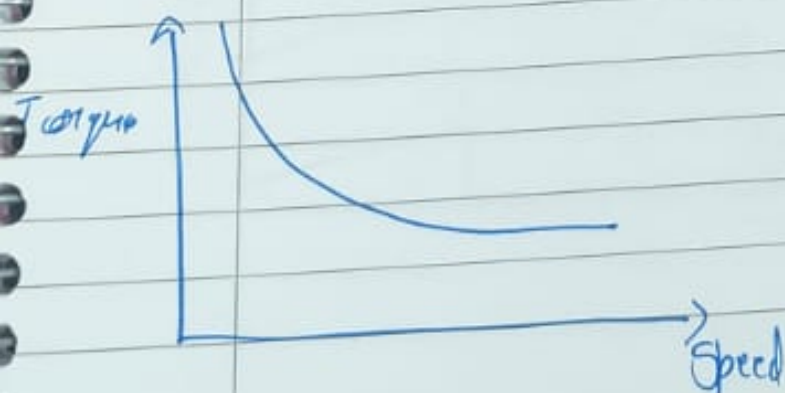


$$\phi = f(I_a)$$

$$\uparrow T = K_e \phi I_a \uparrow \quad (\text{This provides high torque for high speed})$$

$$T = \frac{K_f K_e V_a^2}{(R_a + K_e K_f)^2} \quad T = \frac{K_f K_e V_a^2}{(R_a + K_e K_f \cos \alpha)^2} \quad (\text{parabolic})$$

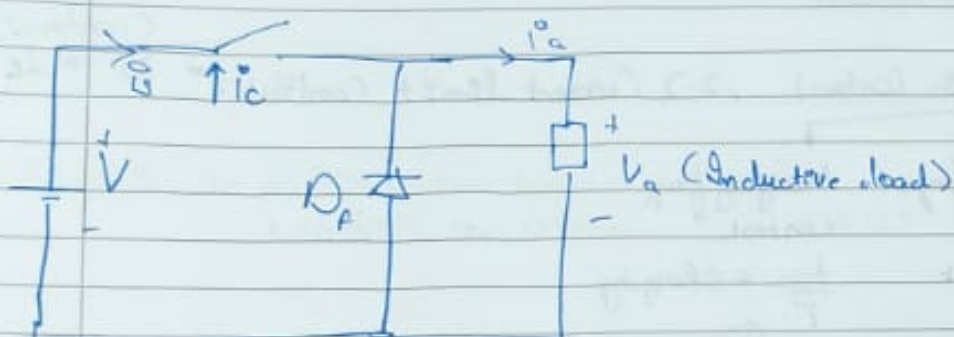
Its disadvantage is when load is less speed is very high.



Chopper Control for DC Motors:-

Advantages are:-

- High efficiency
- Light weight
- Quick response
- flexibility in control
- chopper $\rightarrow$ high frequency operation $\rightarrow$ high less ripple $\leftarrow$ freq. Volt
- lower losses



Control Signal:-

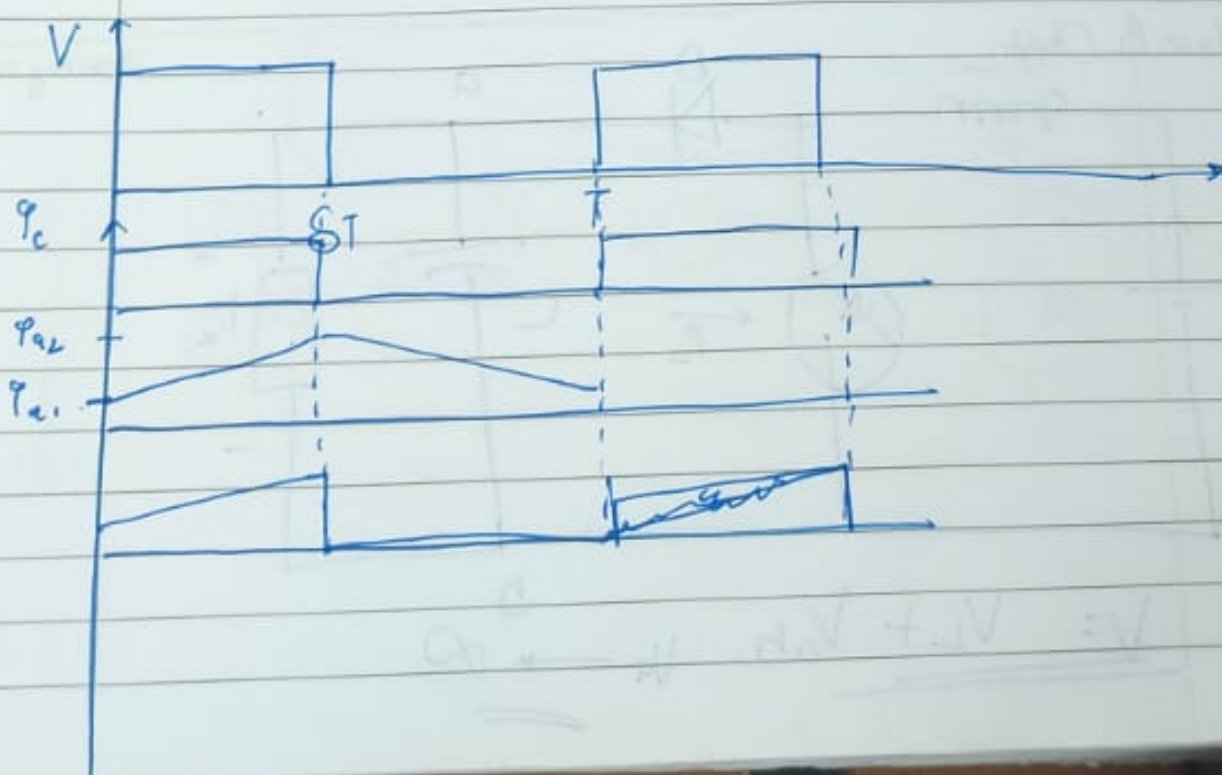
Transistor - Base Current

MOSFET - Gate to Source Voltage

GTO - Gate Current

Thyristor - Gate Current

$$\delta = \text{duty}$$



Diode D_f provides path for load I to flow when switch S is off. It will improve from load current to flow when switch S is off. It will improve current waveform by preventing transient voltage from opposing across switch S .

Types of Control:-

1.) Time Ratio Control:-

Constant freq. Control
 $\frac{t_{on}}{T} = \text{constant}$

Varying freq. control

$\frac{t_{on}}{T} = \text{varying}$

(when current change)

Decrease voltage o/p

2.) Current Limit Control:- $\delta \rightarrow I_L$

(We do not use this much in DC motor control)

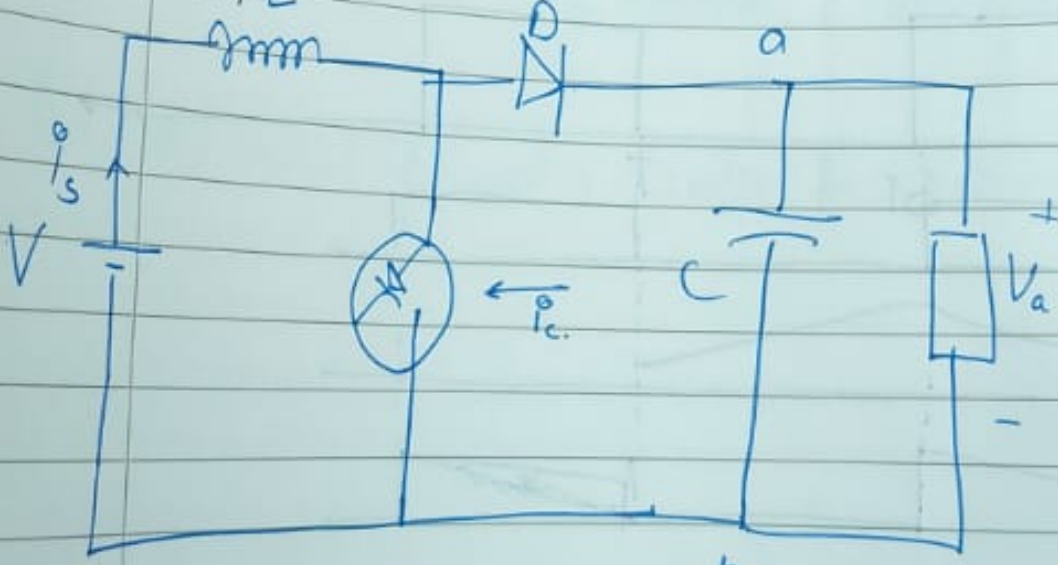
Using motor

no go to lower o/p voltage

This was first quadrant chopper control having +ve speed, torque, current and voltage

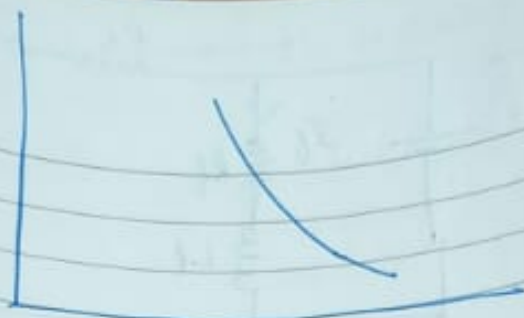
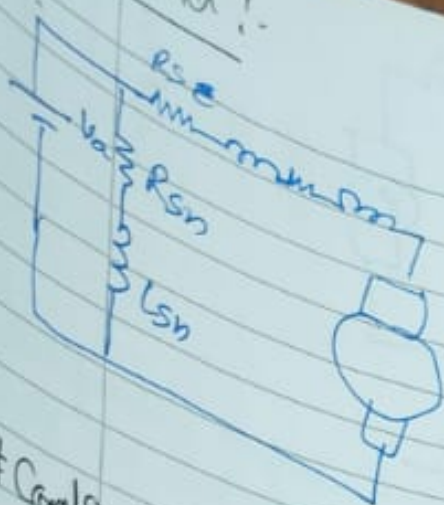
Class B Chopper

31st Aug 21



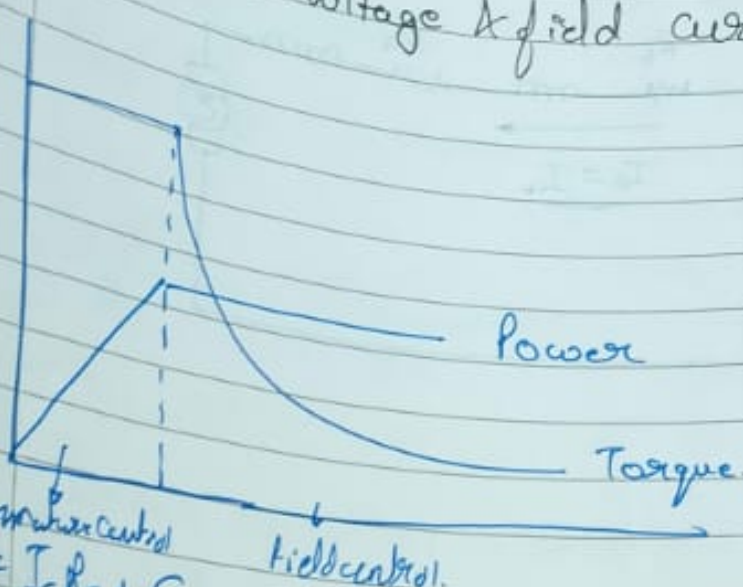
$$V = V_L + V_{ab} \quad V_a \rightarrow \infty$$

Compound :-



(See long and short shunt).

Combined armature voltage & field current control.



Armature control
 $V_a = I_a R_a + E$
 $E = K \phi \omega_m$
 $T = K \phi I_a$

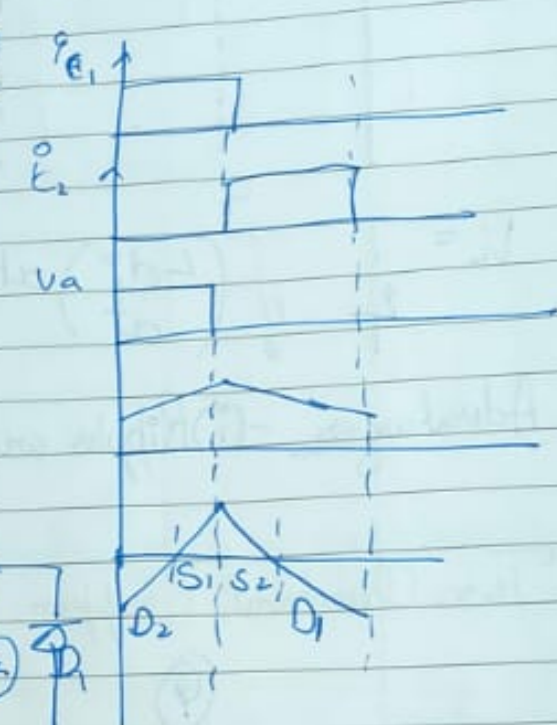
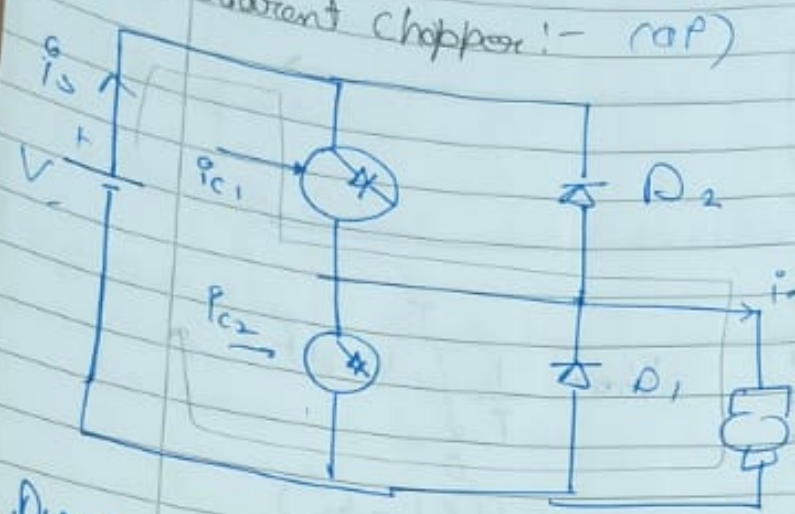
Field control.

For constant torque we use armature voltage control using PE devices. (Choppers)

For field control we get a constant power output

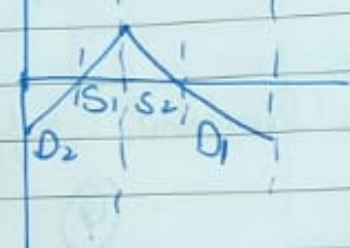
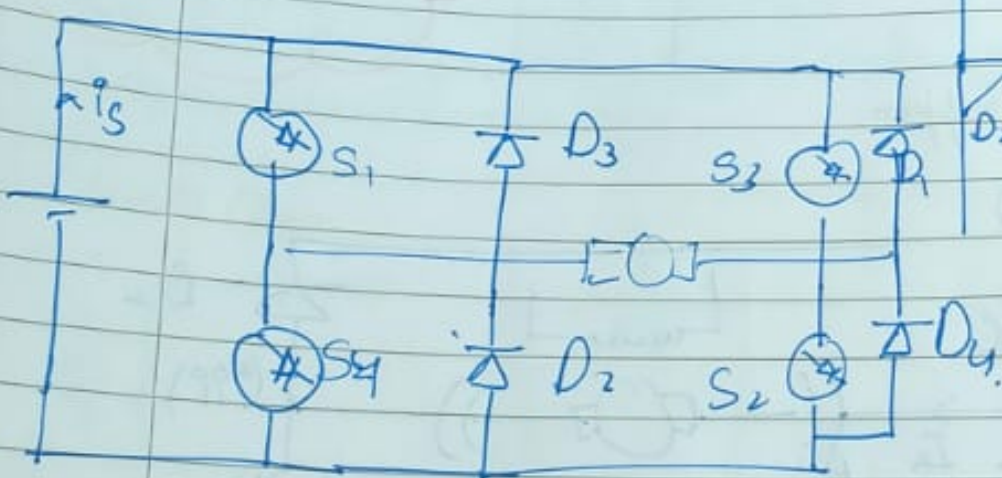
Current will flow from (i) (ii) (iii) (iv) when switch is closed
 When switch is open $\rightarrow$ Current directed through armature
 Remains same but voltage changes. Hence

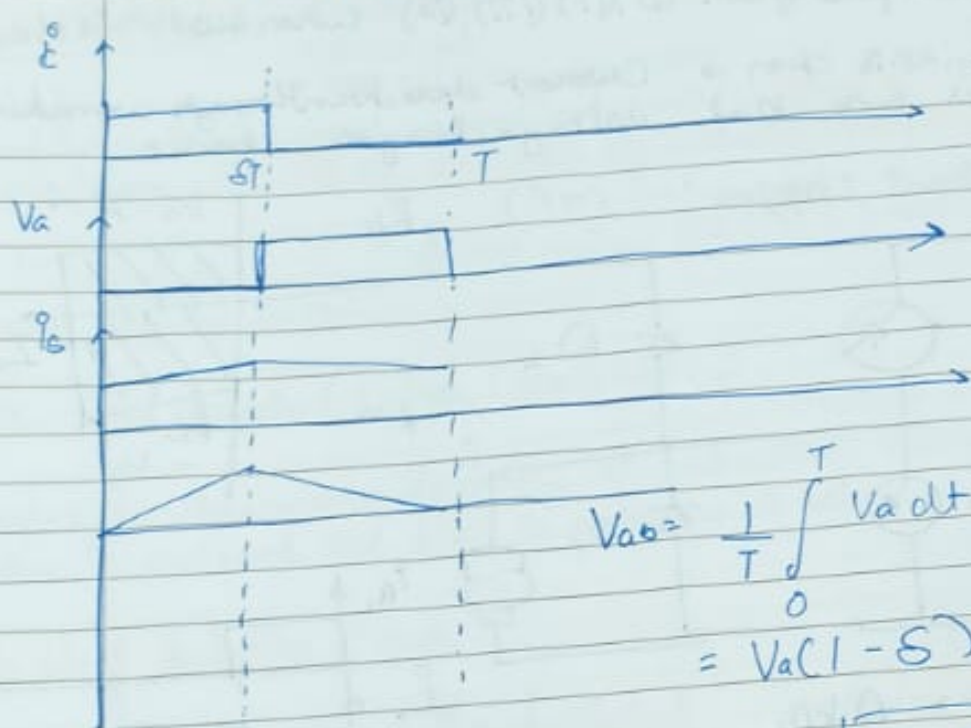
Two Quadrant Chopper:- (a) (b)



During regen $\rightarrow$ D_1, D_2
 During acc. $\rightarrow$ S_1, S_2

Class E four Quadrant



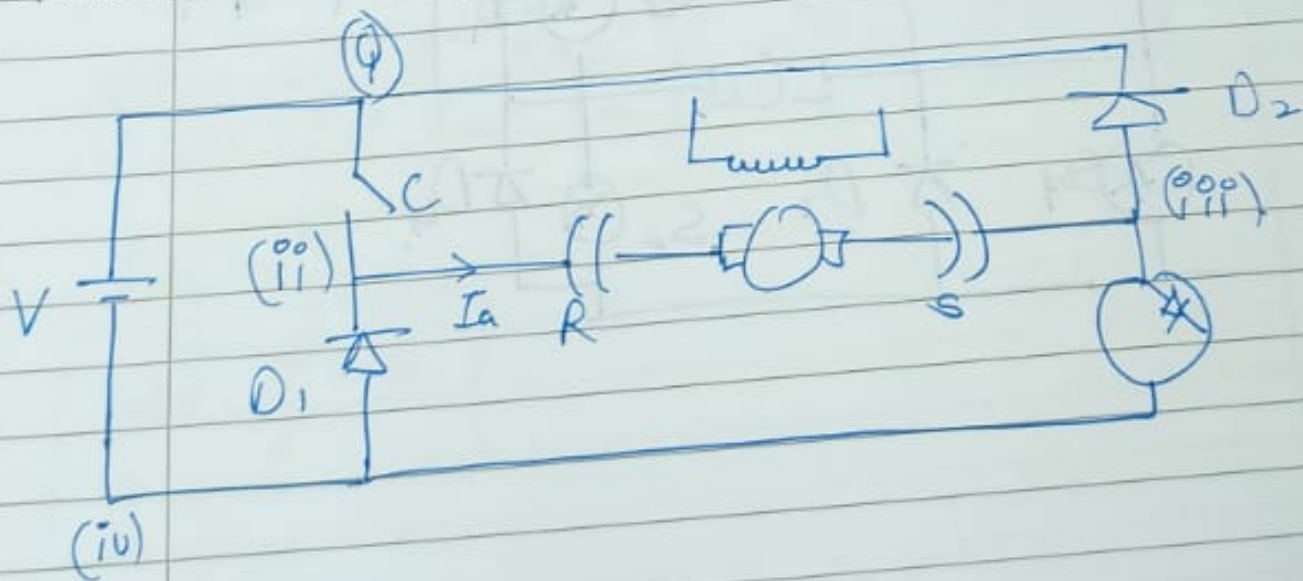


$$V_o = \frac{1}{T} \int_0^T V_a dt = V_a(1-\delta)$$

$$V_L = \frac{1}{T} \int_0^T \left(L \frac{di_L}{dt} \right) dt \rightarrow \underline{0} \quad \boxed{V_L = 0} \text{ average.}$$

Advantages - (i) Ripples are less in source voltage. $V_a = \frac{V}{(1-\delta)}$

Two Quadrant Chopper

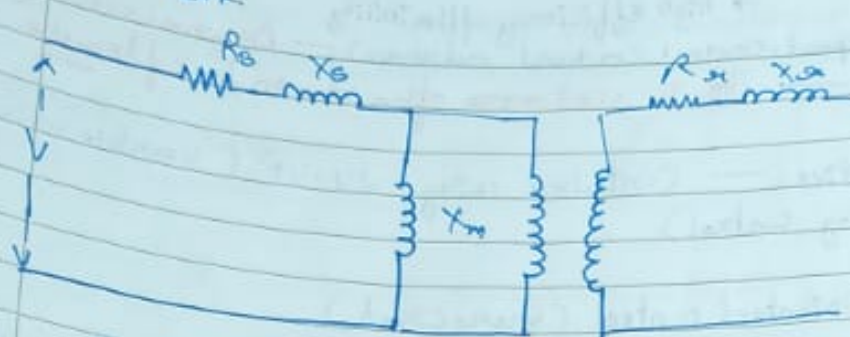


Steady State Performance

2/07/2020

Eq. (1)

Stator - L & R



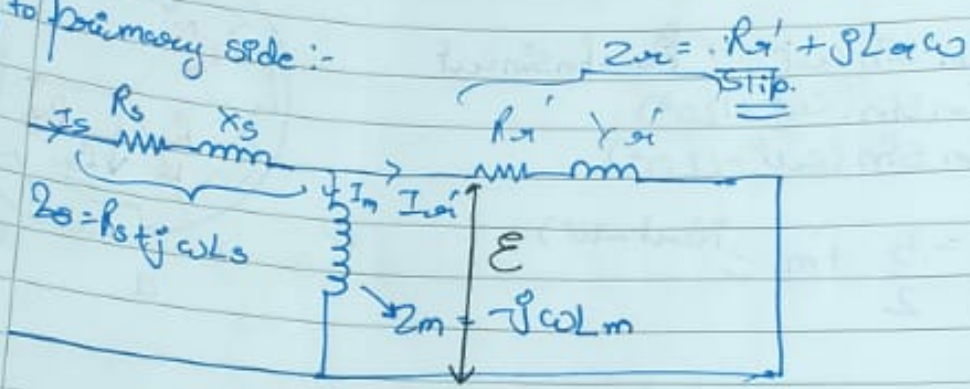
Equations:

$$a_{11} = \frac{R_s}{R_r}$$

for squirrel cage motor

$$(n_s = 1)$$

Referring to primary side:-



$$I_s = \frac{V}{Z}, \quad I_a' = \frac{Z_m}{Z_m + Z'_r} I_s$$

$$R'_r = a_{11}^2 R_r$$

$$X'_r = a_{11}^2 X_r$$

$$\text{Power} = 3 I_a'^2 \frac{R'_r}{s}$$

$$T = \frac{P_{mech}}{\omega}$$

Revolving Magnetic field rotates at Synchronous speed

ω_{ms} = Synchronous Speed

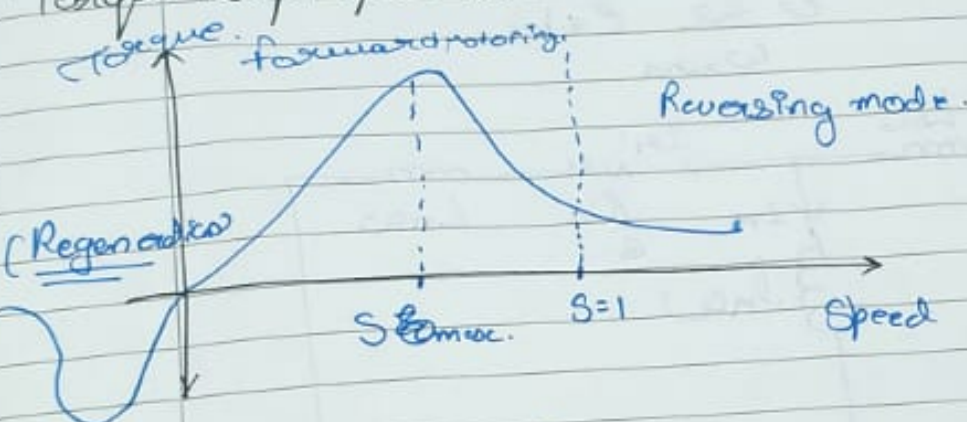
ω_{rs} = Rotor Speed

ω_{sl} = Slip Speed

$$\omega_{sl} = \omega_{ms} - \omega_{rs}$$

$$s = \frac{\omega_{ms} - \omega_{rs}}{\omega_{ms}} = \frac{\omega_{sl}}{\omega_{ms}}$$

Torque Slip / Speed Characteristics:



$$\omega_r = \frac{2}{p} \omega (1-s)$$

When $s=1$, $\omega_r = 0$

At $s > 1$ $\omega_r = -ve$ we enter reversing of motor.

When rotor speed $>$ Synchronous speed. $\rightarrow$ Regenerative mode.

Induction Motor:-

- Small size
- low cost
- high efficiency & flexibility

Wound Rotor - Speed Control (external resistance) — Bewick (costly)

Squirrel Cage — This is used more often due to lack of brushes.

→ Cost effective. — Controlled using VV-VF (Variable Voltage - Variable frequency control)

FOC - Field Oriented control (more costly)

Complexity due to non-linearity of Induction motor, VVVF method is not used and FOC is more prominent.

Principle of Induction Motor:-

$$i_{as} = I_m \sin \omega t \rightarrow \phi_{as} = I_m \sin \omega t$$

$$i_{bs} = I_m \sin (\omega t + 120^\circ)$$

$$i_{cs} = I_m \sin (\omega t - 120^\circ)$$

$$i_s = \frac{3}{2} I_m e^{j(\omega t - 90^\circ)}$$

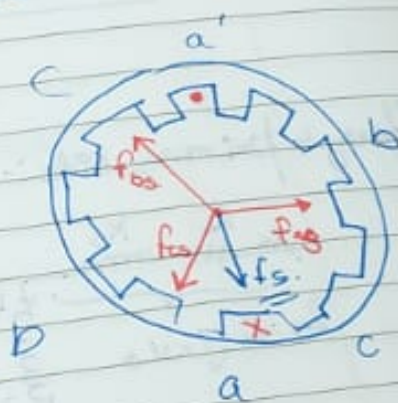
(resultant)

Revolving Magnetic field:-

when $\omega t = 0^\circ$ $i_{as} = 0$, $i_{bs} = -\frac{I_m}{2}$, $i_{cs} = \frac{I_m}{2}$
 at $\omega t = 90^\circ$ $i_{as} = I_m$, $i_{bs} = -\frac{I_m}{2}$, $i_{cs} = -\frac{I_m}{2}$

$$i_s = 0$$

$$\omega_m = \frac{p}{2} \omega_e$$



at max torque $\rightarrow \frac{dT}{dR'_s} = 0$

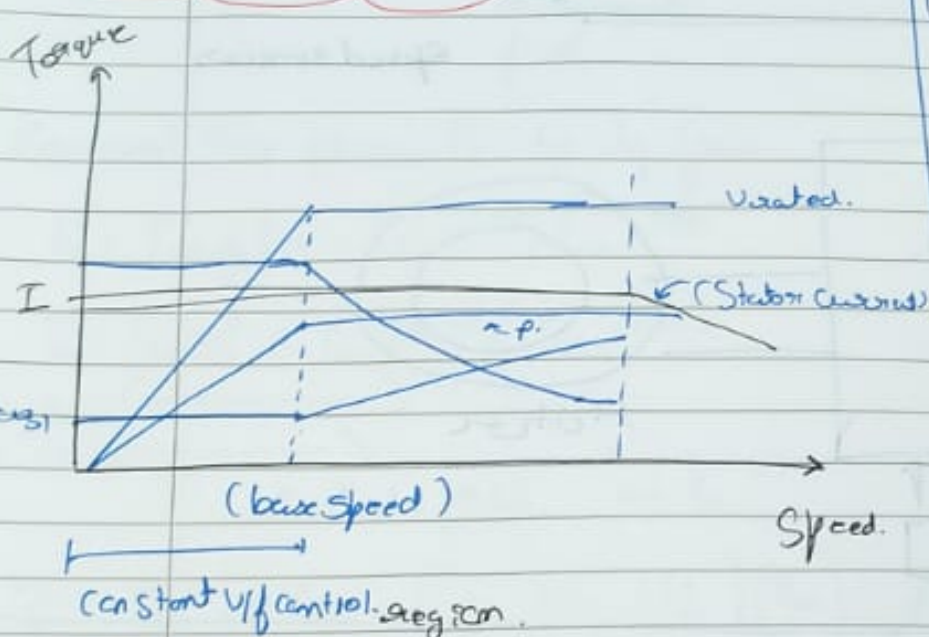
$$\frac{R'_s}{s} = \omega L_s$$

$$s = \frac{+R'_s}{\omega L_s}$$

sl: for max torque

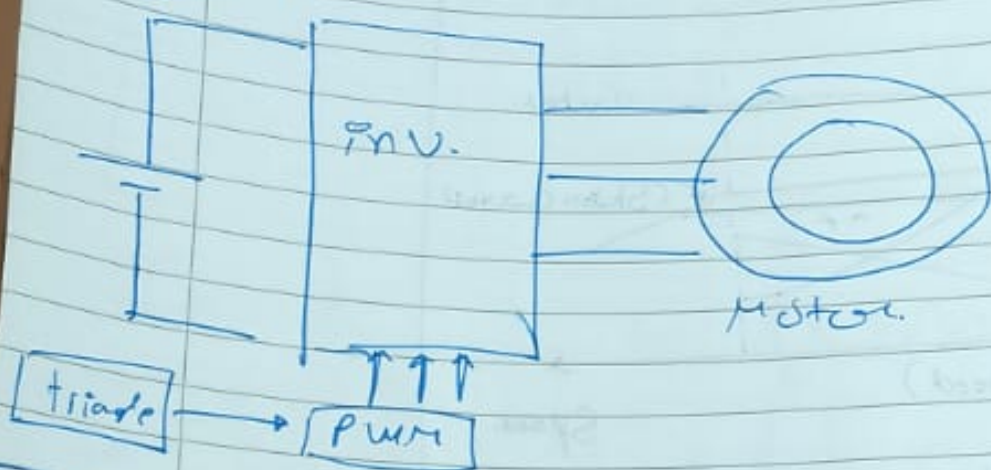
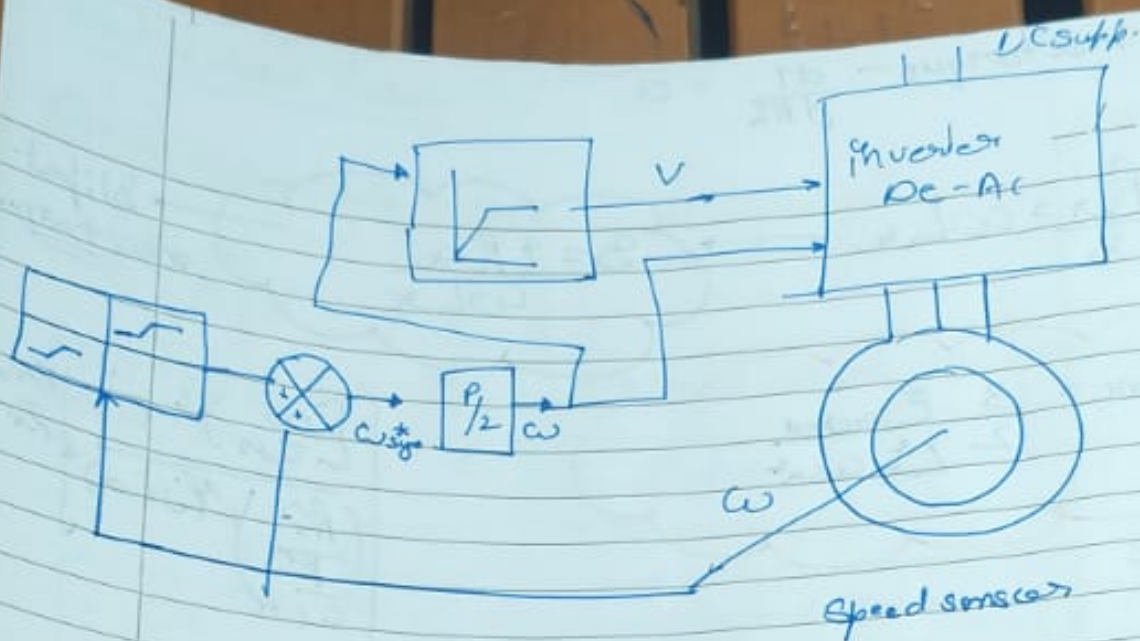
$$T_{max} = \frac{3}{2} \frac{E_{rated}^2}{L_s \omega_{rated}}$$

$$\left[\frac{\omega}{\omega_{rated}} \right]^2 \frac{E_{rated}^2}{\left(\frac{R'_s}{s} + \omega L_s \right)^2}$$

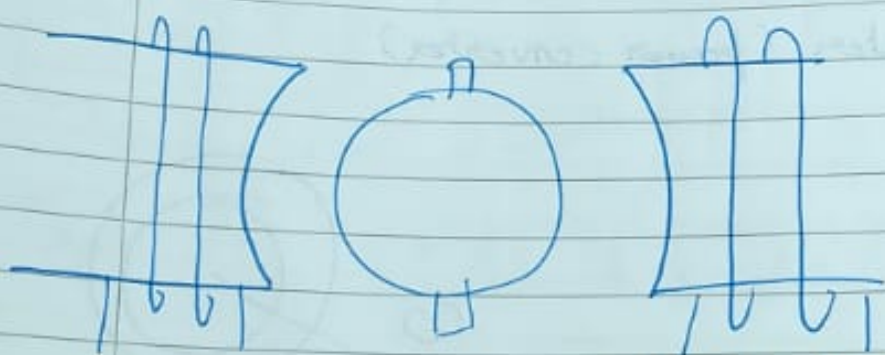


To control $V \rightarrow$ inverter (power converter)





8th Sept. 2026



Speed control

DC Motor two current $\rightarrow I_f, K I_a$ (control using only)

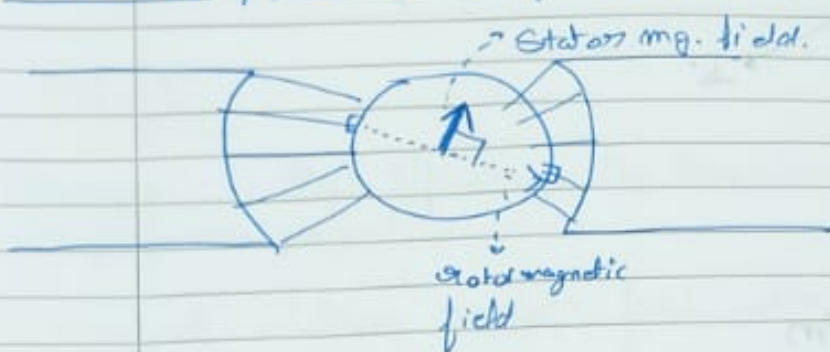
$P = K_f I_a$ (Torque control)

for Induction Motor.

$\rightarrow$ We only have one controlling quantity $\rightarrow$

Stator Current $\rightarrow$ Non linear behaviour - Open loop control using V/f control.

For closed loop control $\rightarrow$ field control in Ind. motor.



Convert three phase AC to dq frame.

$$f_s = f_{as} \cos \theta + f_{bs} \sin \theta + f_{cs} \cos \theta$$

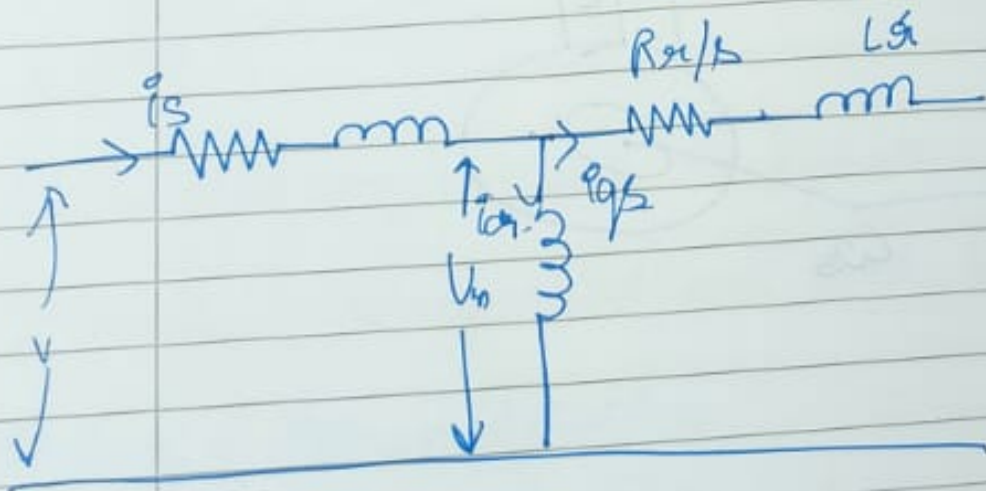
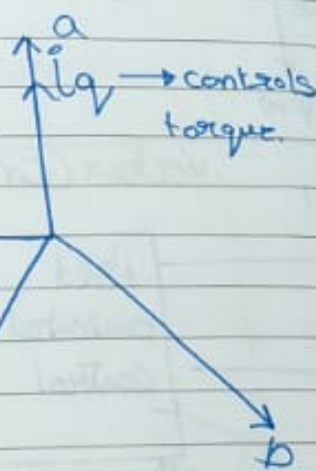
$$i_s = i_{as} \cos \theta + i_{bs} \sin \theta + i_{cs} \cos \theta$$

$$v_s = (\quad)$$

$$i_s = (\quad)$$

controls field. $\leftarrow i_d$

$$i_s = i_{qs} + j i_{ds} = \begin{bmatrix} 1 & 1/2 & 1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \\ 0 & -\sqrt{3}/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix}$$



Induction Motor Control

7/07/2022

Constant Voltage / Hz control

$$E \propto \phi$$

$$E = k \phi \omega$$

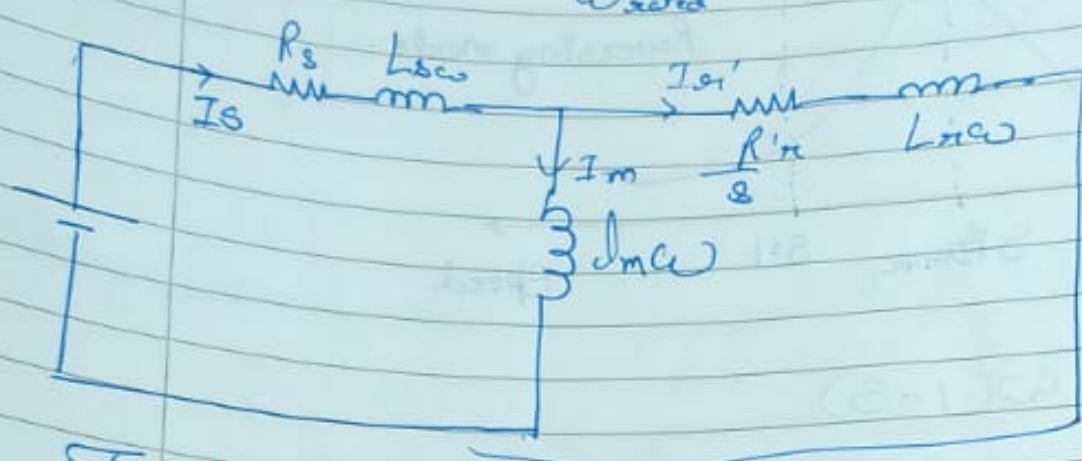
$$250V \rightarrow 50 \text{ Hz}$$

$$125V \rightarrow 25 \text{ Hz}$$

$$E = k \phi$$

Keep ϕ constant the Torque = constant

$$E = \frac{\omega}{\omega_{rated}} E_{rated}$$



$$I_{r'} = \frac{E}{Z_r}$$

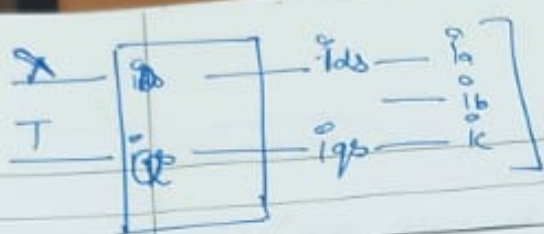
$$Z_r = \frac{R'_r}{s} + j\omega L_{r\omega}$$

$$I_{r'} = \frac{\omega}{\omega_{rated}} \frac{E_{rated}}{\frac{R'_r}{s} + j\omega L_{r\omega}}$$

$$\frac{R'_r}{s}$$

$$Torque = \frac{P}{\omega} \rightarrow P = 3 I_{r'}^2 \frac{R'_r}{s}$$

$$= 3 \left[\frac{(\omega/\omega_{rated})^2 E_{rated}^2}{\left(\frac{R'_r}{s}\right)^2 + (\omega L_{r\omega})^2} \right] \frac{R'_r}{s}$$



from Inverter terminals

BLDC

Advantages: 1) High efficiency - No brush losses, copper-rotor loss
uses permanent magnets.

2) Compact: Strong magnetic field with small space.

3) Easy to control — Use of inverter $(T \propto \phi I_a) - DC$
 $T \propto I_a (\phi = \text{constant})$

4) Ease of cooling - No heat loss in stator wdg. - easier to cool.

5) Low maintenance.

6) Low noise emission.

Disadvantages: 1) Higher cost due to perm. mag.

2) Limited constant power range.

$$E \propto \phi \omega$$

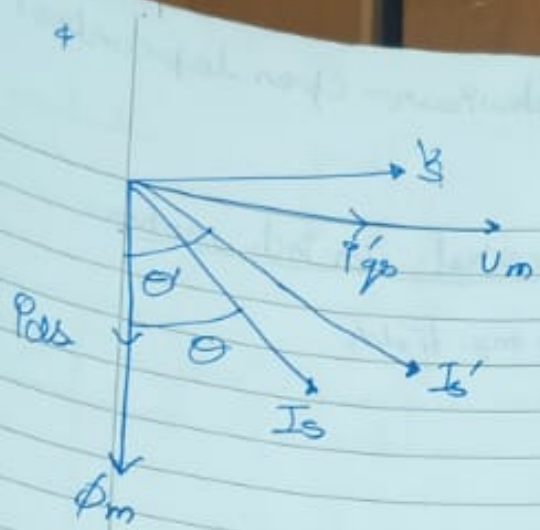
can only run at twice the base speed.

$\omega \propto \frac{1}{\phi}$

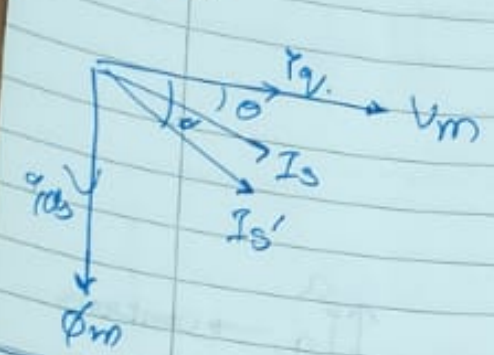
Due to field being constant the power output is limited.

3) Safety - demagnetisation and locking

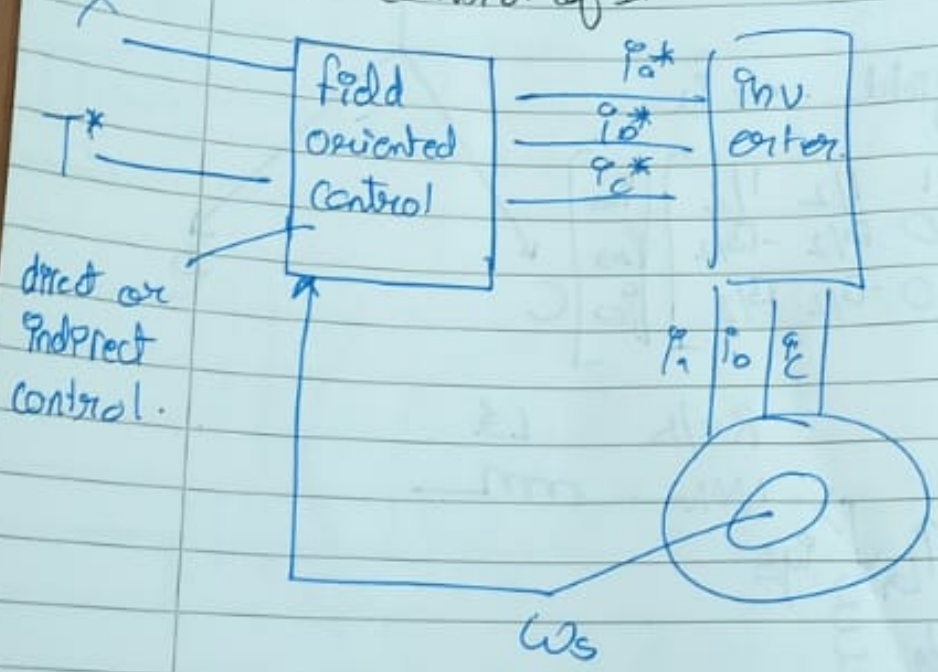
Magnet-demagnetisation -



Keep flux constant to avoid
change core saturation angle to
control speed/torque



Vector Control of IM



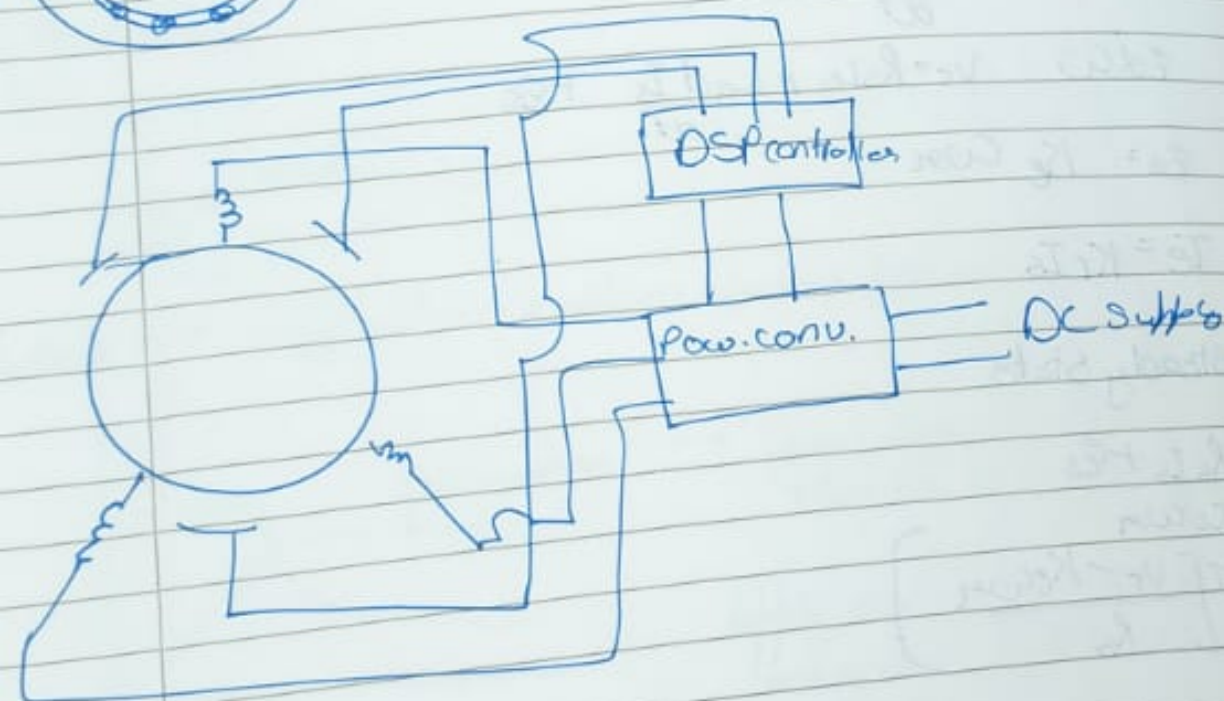
direct or
indirect
control.



(120° +ve, +120° -ve, 60-60 empty)

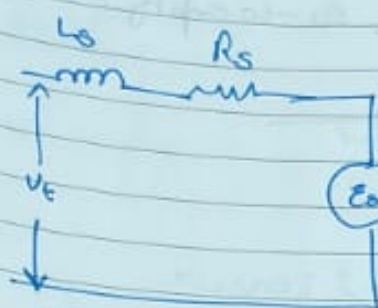


Hall effect sensor



Performance Analysis of BLDC

Torque Speed Characteristics $\rightarrow$ field & current $\rightarrow$ constant.



At steady state

$$T_e = T_L + J \frac{d\omega}{dt} + B\omega$$

$$V_t = R_s I_s + L_s \frac{dI_s}{dt} + E_s$$

$$E_s = K_e \omega$$

$$T_e = K_T I_s$$

At steady state

$$V_t = R_s I_s + E_s$$

$$E_s = K_e \omega$$

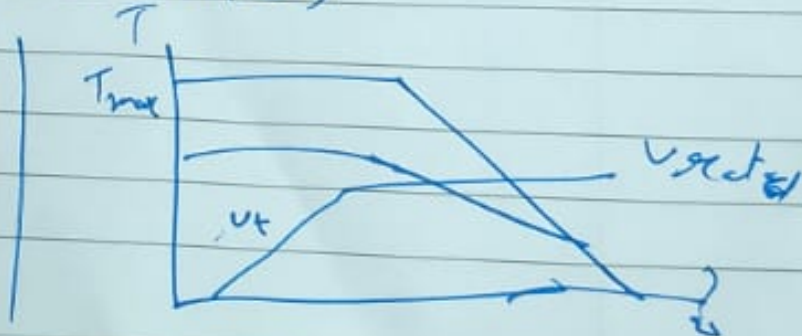
$$T_e = K_T \left[\frac{V_t - K_e \omega}{R_s} \right]$$

$$T_e(s) = T_L(s) + J s \omega(s) + B \omega(s)$$

$$V_t(s) = R_s I_s(s) + L_s s(I(s)) + E_s(s)$$

$$E_s(s) = K_e \omega(s)$$

$$T_e(s) = K_T I_s(s)$$



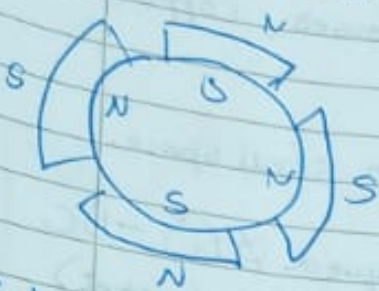
BLDC:-

Stator:-

Rotor - Perm. Magnet

Commutator - Electronic

On the basis of stator:-



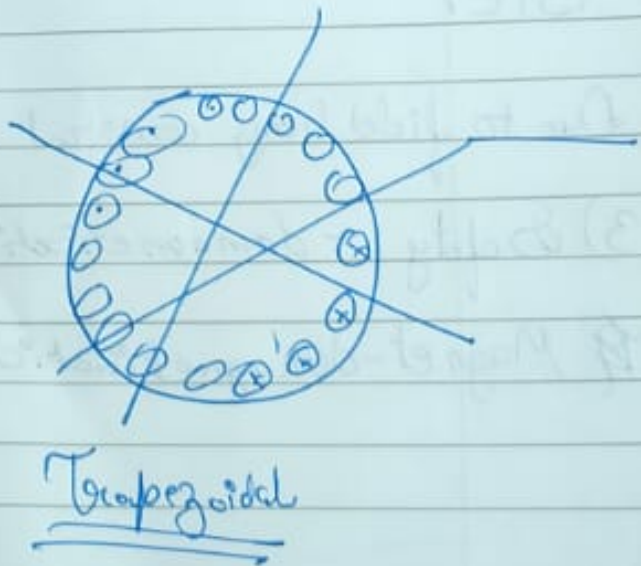
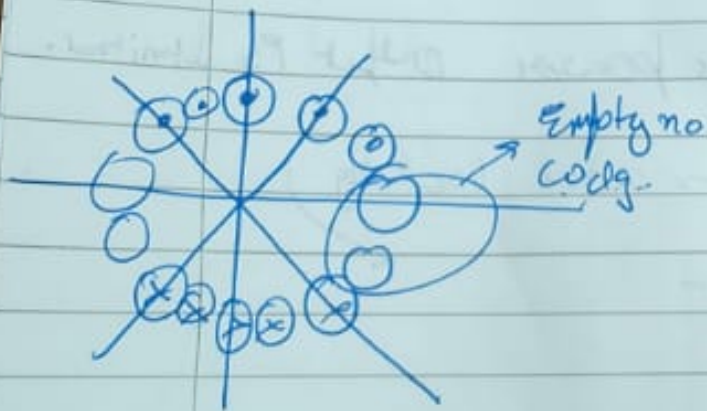
External/Surface
mounted stator



Interior mounted



- Minimise Cogging torque
- Cannot use for high speed app.



Switch Reluctance Motor:- (SRM)

10/04/2020

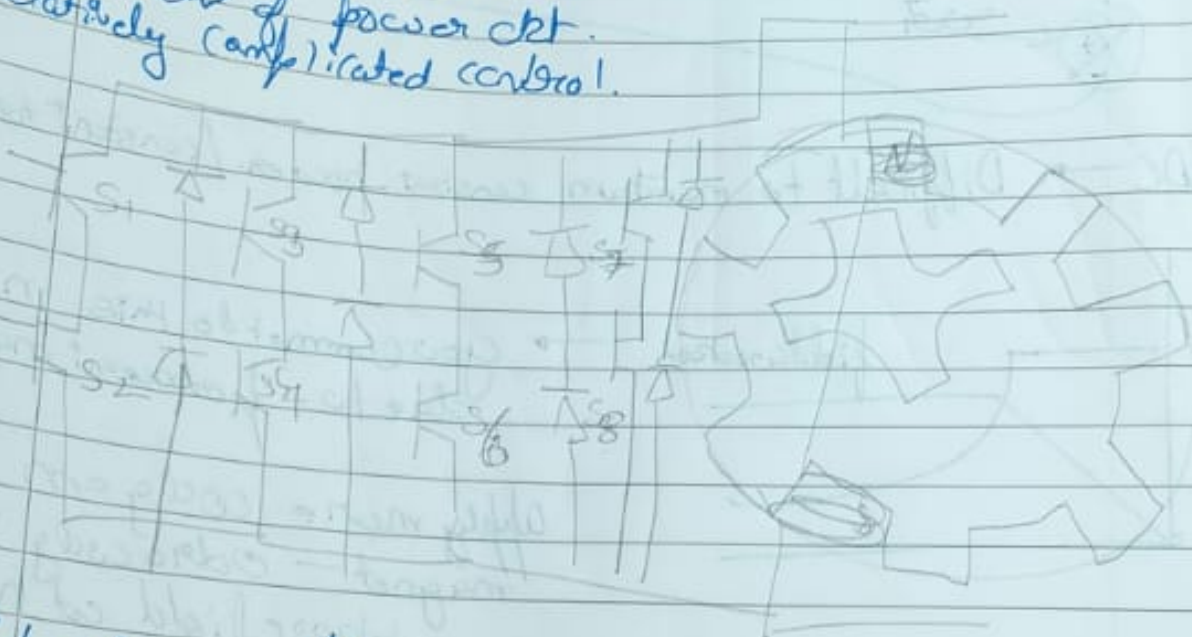
$$I = V/R, \phi = \frac{NI}{R}$$

- Simple Rotor Construction
- Rugged mechanical "
- Absence of stator cogs & Perm. magnet
- Good fault tolerance
- Suitable for wide range of operation

Rotor torque is produced when rotor aligned with d.c.c reluctance position.

Disadvantage:-

- Ripple torque.
- Acoustic noise
- Non linear characteristics
- Requirement of power ckt.
- Relatively complicated control.

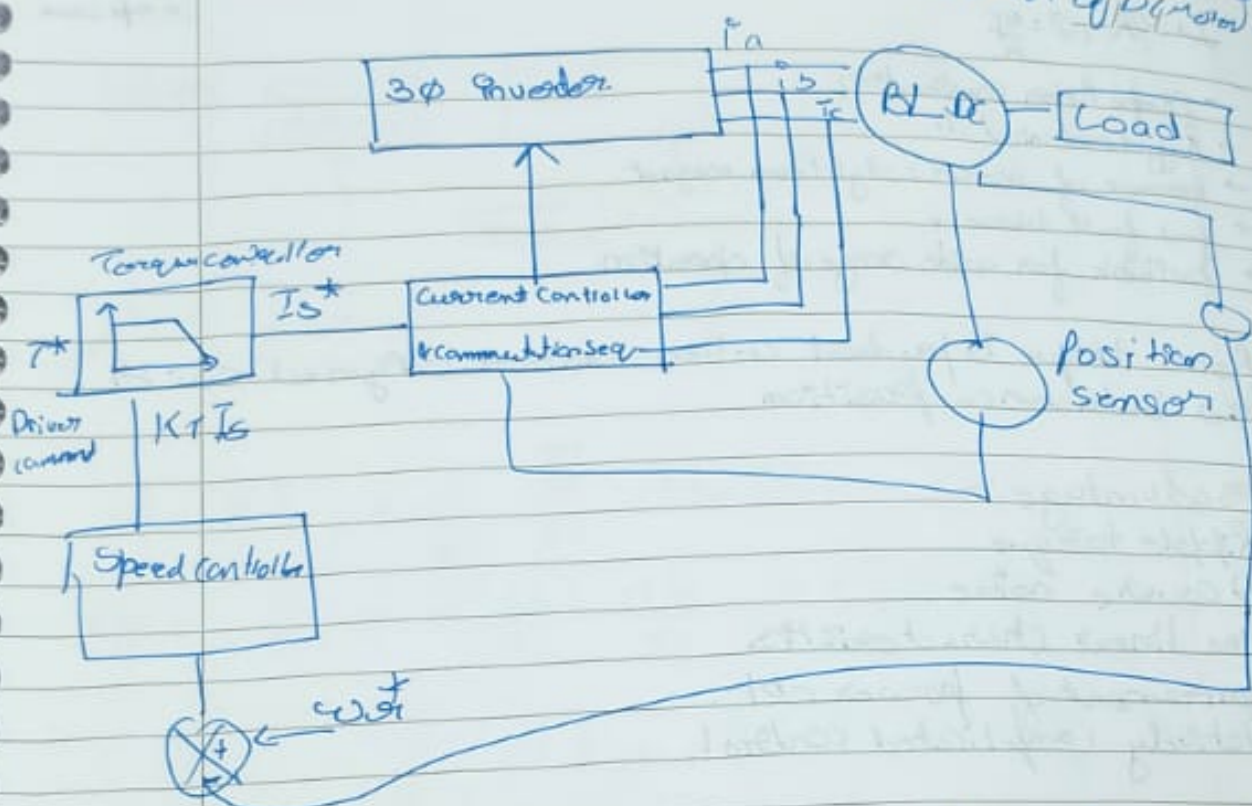


$T \propto \frac{dL}{d\theta}$ How L changes

Unaligned	Aligned.
R_{max}	R_{min}
L_{min}	L_{max}
flux linkage min	flux linkage max.

Performance Analysis of BLDC:-

(see detail / indexing control of DCMotor)



In BLDC → Difficult to maintain constant power / constant torque.



fieldweaken

you cannot do this in BLDC due to permanent magnets.

Apply more ω & ω on permanent magnet - extra ω is applied field at higher speeds.

$$= Ri + \sum_{k=1}^m \left\{ \frac{\partial (L_{pk} i_k)}{\partial i_k} \cdot \frac{di_k}{dt} + \frac{\partial L_{pk}}{\partial \theta} \frac{d\theta}{dt} \right\}$$

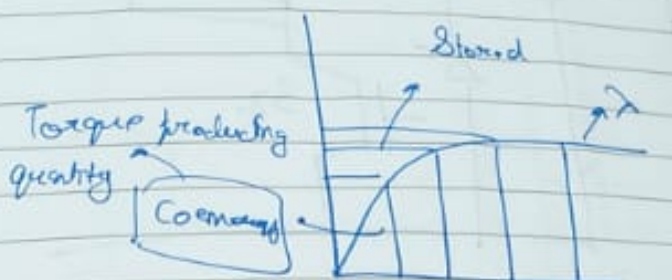
$$= Ri + \sum_{k=1}^m \left[L_{pk} + i_k \frac{dL_{pk}}{di_k} \right] \frac{di_k}{dt} + i_k \frac{dL_{pk}}{d\theta} \omega$$

$$= V = Ri + L_{ss} + \sum i_k \frac{dL_{pk}}{di_k} \frac{di_k}{dt} + \sum i_k \frac{dL_{pk}}{d\theta} \omega$$

Torque Production:

$$W_f = \int_0^{\theta} \lambda di$$

$$\text{Torque} = \left[\frac{dW_f}{d\theta} \right]_{i \text{ constant}}$$



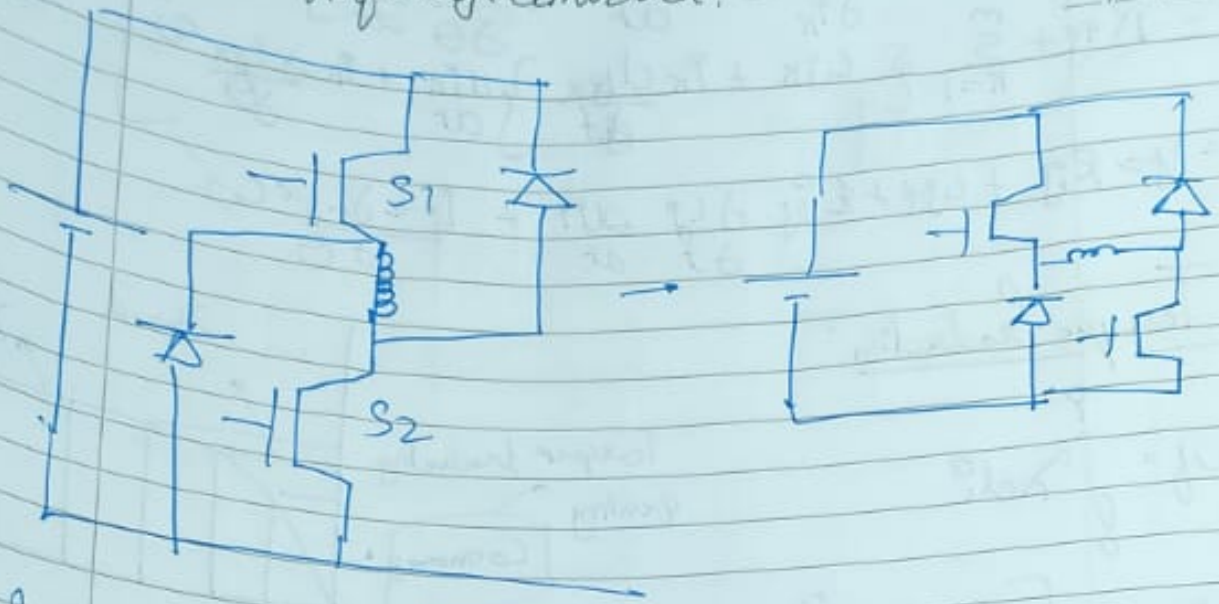
$\lambda \neq L i$ for unsaturated $\Rightarrow \lambda = L i$

$$W_f = \frac{1}{2} L i^2$$

$$T = \frac{1}{2} i^2 \frac{dL}{d\theta}$$

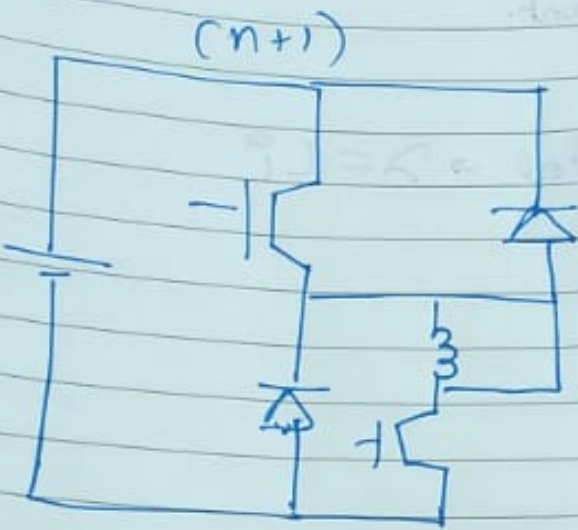


1) Classical half bridge converter:-

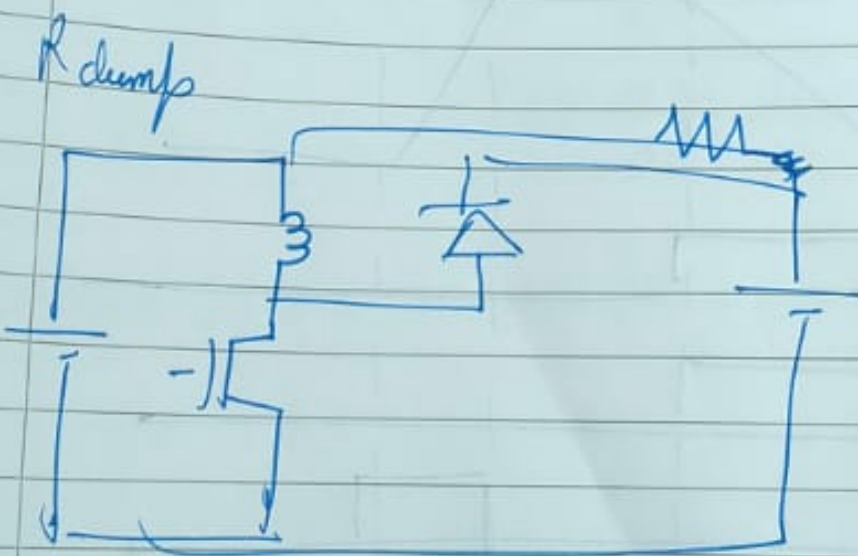


~~Snell c~~

(2)

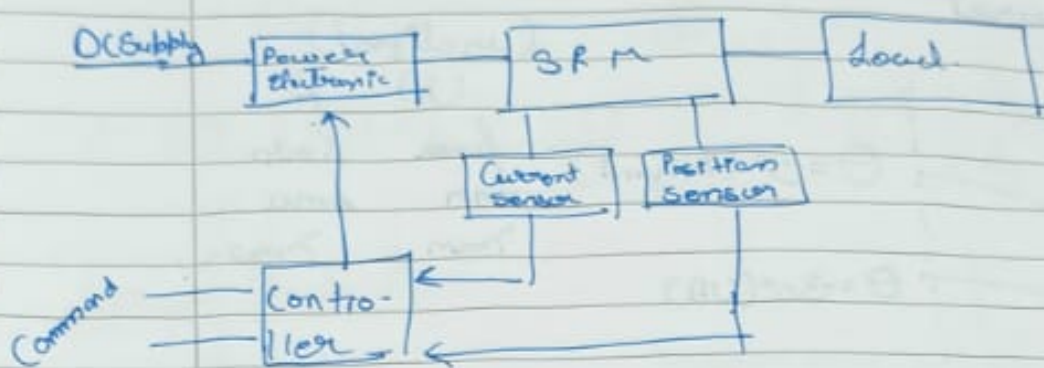


(3)



Overcurrent

$$L = \frac{\lambda}{i} \rightarrow L = \lambda(i, \theta)$$



$$V_i = R i_i + \frac{d}{dt} \sum_{k=1}^m \lambda_{ik} i_k$$

$$= R i_i + \left\{ \sum_{k=1}^m \left[\frac{\partial (L_{ik})}{\partial i_k} \cdot \frac{d i_k}{dt} + \frac{\partial (L_{ik})}{\partial \theta} \frac{d \theta}{dt} \right] \right\}$$

$$= R i_{\text{pm}} + \underbrace{\sum_{k=1}^m \left[L_{ik} + i_k \frac{d i_k}{dt} \right]}_{\text{Inductance}} + \underbrace{\frac{\partial L_{ik}}{\partial \theta} \omega}_{\text{back emf}}$$

Rotor resistance
Inductance
back emf.

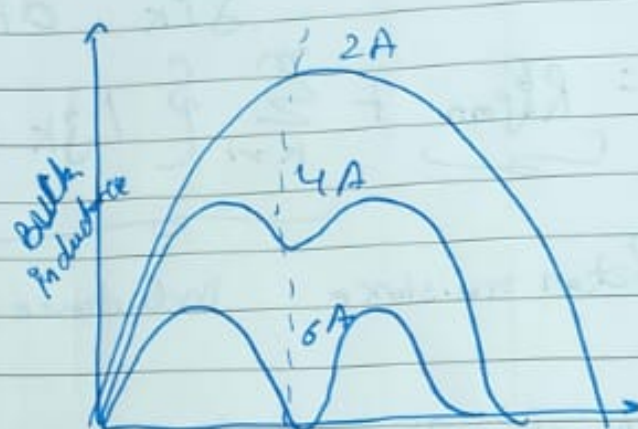
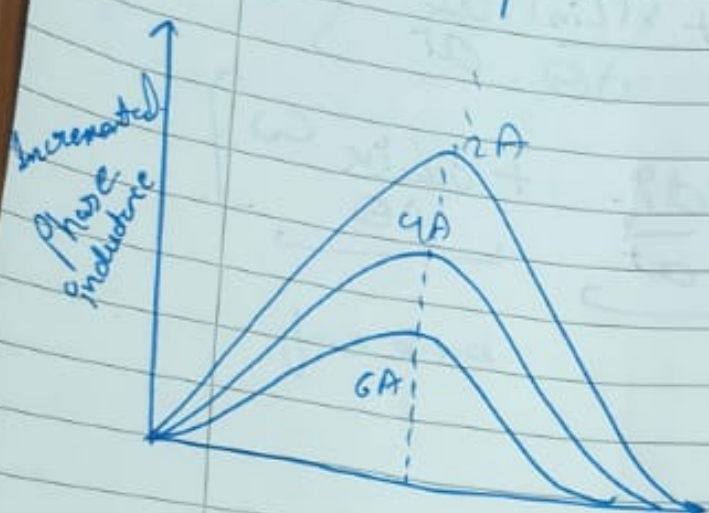
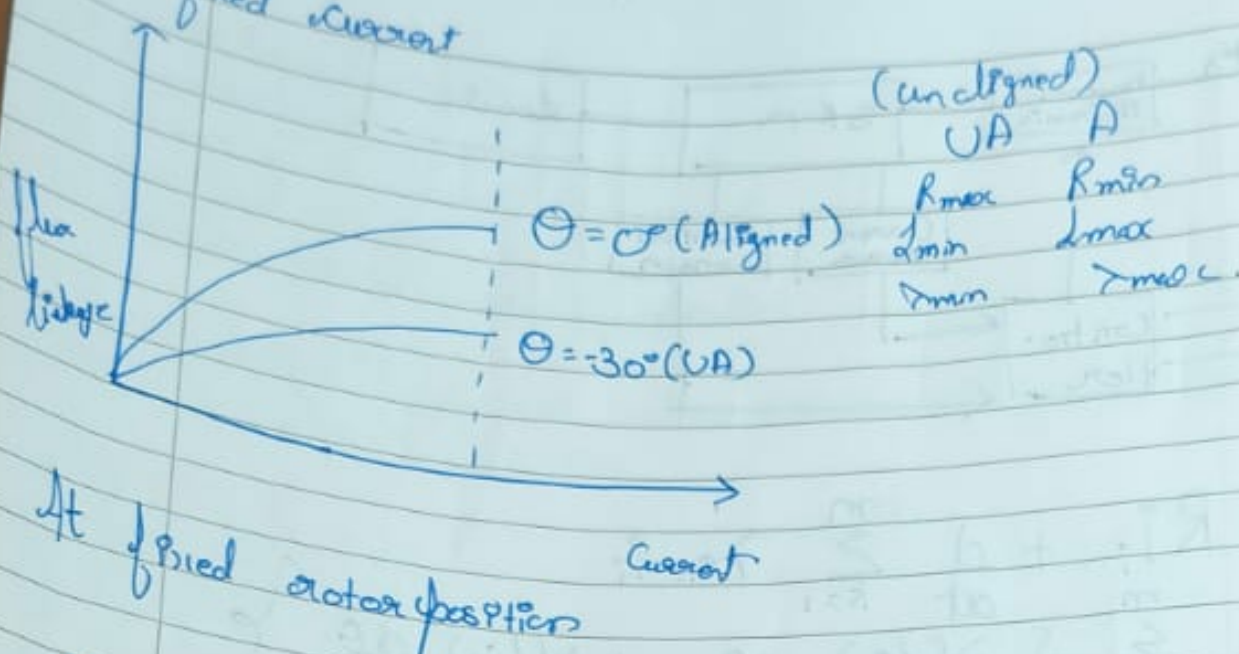
~~bulk~~ Bulk Inductance.

$$L = \frac{\lambda}{i}, \quad L_{enc} = \frac{\partial \lambda}{\partial i}$$

$$T_{ij} = \frac{d}{dt} L_{ij}(\theta, \dot{\theta}) \dot{\theta}$$

fixed current

21 Apr 2024



$$V_i = R_{ip} + \frac{d}{dt} \sum_{k=1}^m \lambda_{ik}$$

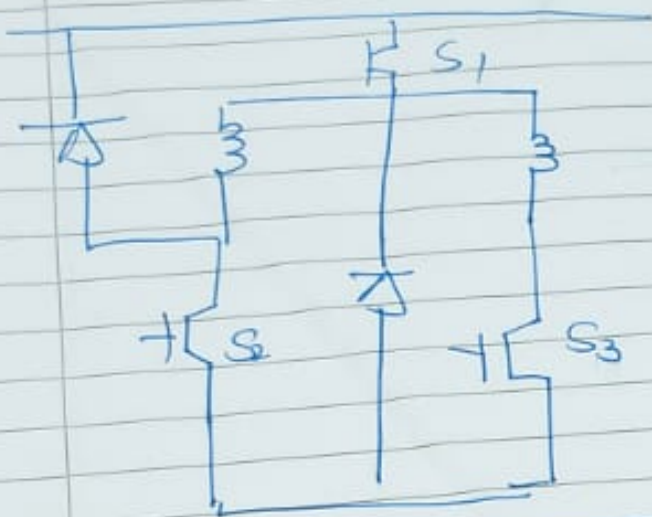
Rotational angular

$$= R_{ip} + \sum_{k=1}^m \left[\frac{\partial \lambda_{ik}}{\partial i_k} \frac{di_k}{dt} + \frac{\partial \lambda_{ik}}{\partial \theta} \frac{d\theta}{dt} \right]$$

$$= R_{ip} + \sum_{k=1}^m \left[L_{ik} + i_k \frac{\partial L_{ik}}{\partial i_k} \right]$$

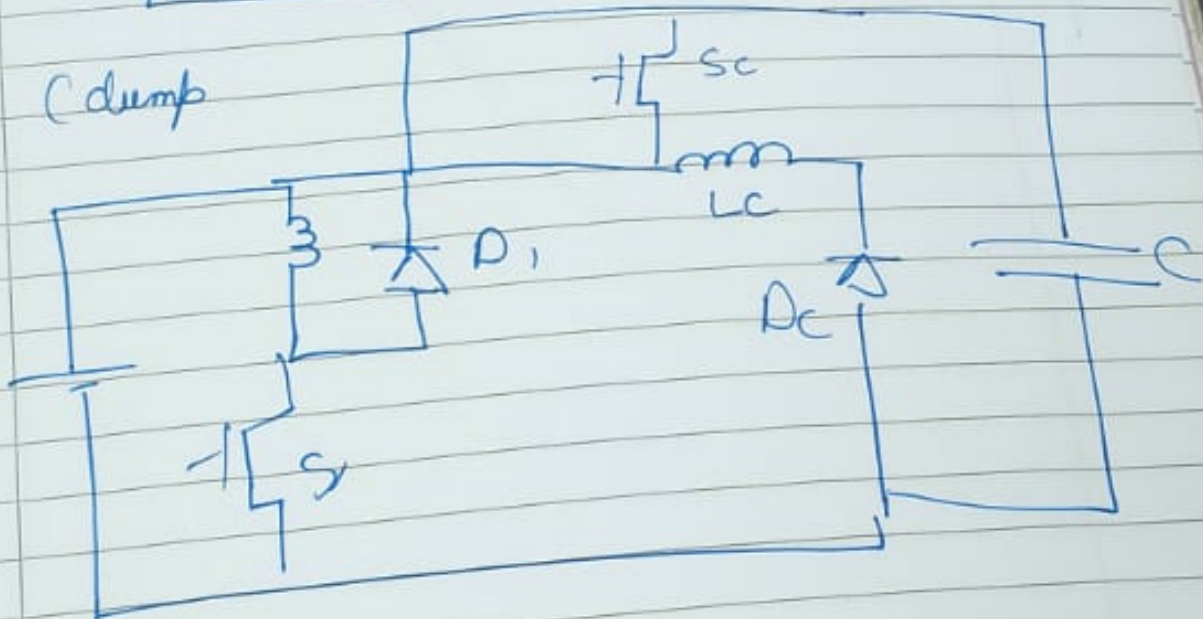
④

1-5n



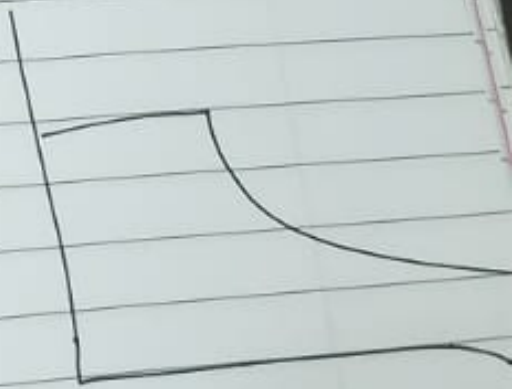
⑤

(dump)



Mode of operation

Back EMF = DC bus voltage





Back to 1/4 = 1/4 from nothing

